Entropic attractor networks: a scale-agnostic, first-principles route to inference and learning in brains and machines

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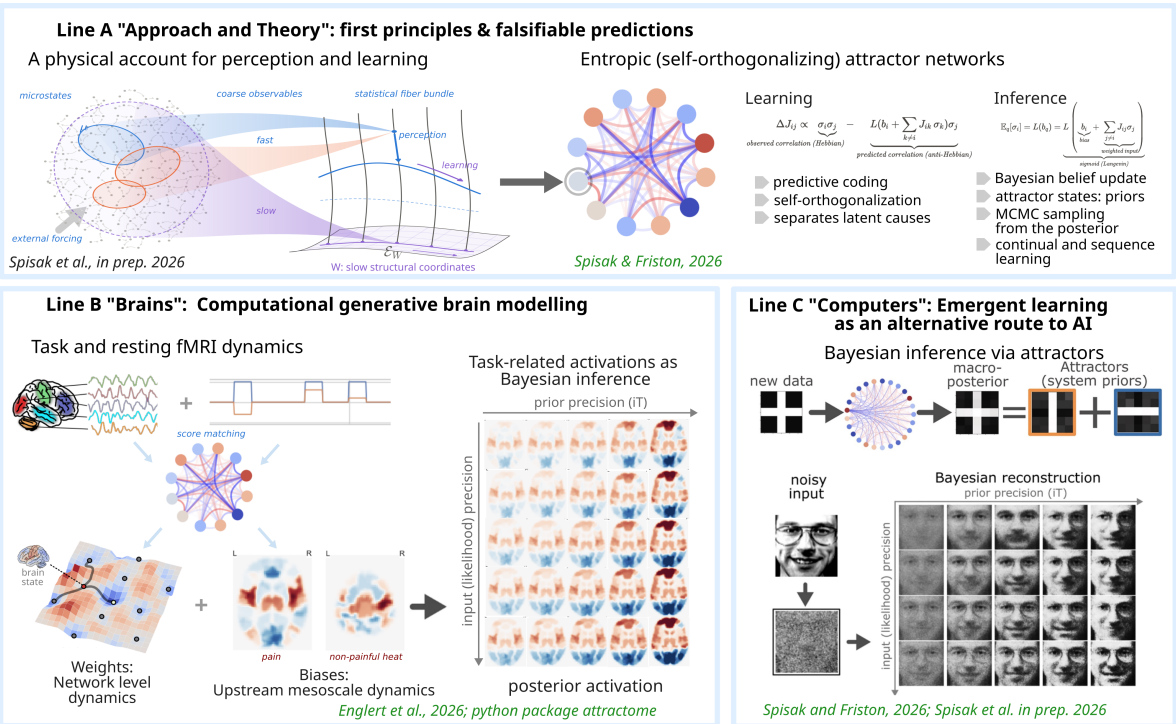
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Abstract

Human brains learn continuously without erasing memories, represent uncertainty, and act on internal world models, all while running on roughly 20 watts. Current AI still struggles to match these capabilities. Neuroscience closely maps the underlying mechanisms but lacks a unifying framework that bridges disciplines and scales. A shared framework would benefit both fields: supplying AI with foundational design principles and giving neuroscience testable, scale-independent accounts applicable to brains and machines alike. This paper outlines a research programme built around such a framework. It posits that basic physical principles of inference and learning govern neural mechanisms, with brains merely refining their exploitation rather than inventing new rules. From this perspective, cognition is not an algorithm imposed upon matter but the repeated, multi-scale utilization of appropriately organized matter's inherent capacity to infer and learn. Central to this effort (Line A) is a formal theory of this capacity and a constructive path toward the emergent collective dynamics, called entropic attractor networks (eANNs). Rather than hand-crafted, eANNs emerge naturally, functioning as both generative models of brain dynamics and trainable stochastic networks. Early results support both roles. This enables us to validate and deploy the theory in the study of biological brains (Line B), harness its emergent properties in AI (Line C), and unify both domains under a single explanatory lens. Confirming our hypothesis would explain why key neural mechanisms take their observed forms and chart a principled route to human-like artificial intelligence.



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About this document: its origins, status and how to contribute
This white paper presents a research vision still awaiting full formalization and testing. It draws on years of empirical, simulation, and theoretical work, some published and much still in preparation. Together, these findings suggest a coherent account of how inference and learning emerge from appropriately structured matter, and how clarifying this process could advance both neuroscience and AI. The goal is to state this framework explicitly, enabling rigorous development and systematic testing.

Rather than a finished theory, what follows is a *research programme* in the sense of Lakatos (1978). It rests on a small set of core assumptions (Section 3), revisable auxiliary hypotheses, and falsifiable predictions (Section 7). Its primary value lies in the testable claims it generates and the research directions it structures, not in current explanatory breadth. We expect several ideas to ultimately prove incorrect, and have framed each one so that even its falsification would yield useful insights.

This document will be version-controlled and updated as new results arrive. Feedback, criticism, and collaboration are welcome. Substantive objections will be recorded and addressed in later drafts. Contributors and advisors who shaped earlier versions will be acknowledged with their consent; their inclusion reflects input, not formal endorsement. Responsibility for the content rests with the author.

Executive summary

Brains never stop learning. New information, whether a face, a route, a word - or a neuroscientific theory - is added without erasing what we already know, while tracking uncertainty and acting with intent, based on an internal causal model. Current AI, despite rapid progress, still struggles with these capabilities: large models are trained offline on vast data, often overwrite previous knowledge when learning new tasks, handle uncertainty poorly, and use significant energy. Neuroscience has cataloged many of the mechanisms behind the brain's abilities, from synaptic plasticity to replay and attractor dynamics, but they are not yet tied together by a framework that explains why they work, how they function across scales and how they carry over to artificial forms of cognition.

This white paper outlines a research programme grounded in such a framework. Its core hypothesis is that inference and learning rely on basic physical principles, which brains did not create but came to exploit ever better. The programme translates this hypothesis into a concrete form through a specific class of models, entropic attractor networks (eANNs), alongside a formal theory explaining how and why they infer and learn.

The model family. eANNs are stochastic attractor networks built from simple units. Each unit infers by merging its internal "bias" with weighted input from neighboring units, while each coupling updates locally by contrasting observed correlations against predicted ones (following a predictive-coding rule). Relying solely on these two mechanisms, the networks encode learned patterns as attractors that function as Bayesian priors, push representations toward orthogonality even when inputs are correlated, capture sequences via asymmetric couplings, generalize to unseen data, and avoid catastrophic forgetting (Spisak and Friston, 2026). Four characteristics distinguish this model family:

- **Derived, not designed.** The inference and learning rules arise directly from first principles - a form of physical learning - instead of being tuned for performance. Hopfield networks and Boltzmann machines emerge as special cases.
- **Concrete and plausible.** The rules operate locally and incrementally, noise serves a functional role (encoding precision), and the model relies on a small set of clear parameters: biases and couplings, from which attractors of the system can be extracted directly. These features align well with both biological circuits and emerging thermodynamic hardware technologies.
- **One model, two uses.** A single network serves as both a generative model for large-scale brain dynamics (fit directly to fMRI data) and as a trainable stochastic learner, allowing insights from one domain to transfer to the other.
- **Scale-agnostic and testable.** The architecture is expected to hold up under coarse-graining (Section 4.2), meaning the same model family can describe (or at least approximate) individual cells, local circuits, or entire brain regions. Key signatures - like attractors encoding Bayesian priors, distributed near-orthogonal collective representations, and noise tracking precision — yield concrete, falsifiable predictions with explicit failure criteria (Section 7).

First evidence. The programme does not start from scratch (Section 2). Alongside the analytical and simulation results (Spisak and Friston, 2026), we present initial findings that human functional brain networks exhibit the primary signature of the eANN model family: approximately orthogonal attractors. We also observe that these attractors align closely with canonical large-scale brain networks, replicate across datasets, and capture task-related shifts as well as altered dynamics in clinical conditions (Englert et al., 2026). Independent groups have begun applying eANNs (Watters et al., 2026) and similar, but separately developed attractor landscape approaches (Fasoli et al., 2026; Song et al., 2026; Gosti et al., 2024; Watanabe et al., 2013; Gu et al., 2018; Athanasiadis et al., 2025). Our results are preliminary, but they demonstrate that eANNs are viable, function as intended, and can be plausibly fitted to real brain data.

From matter to units. Our published derivation (Spisak and Friston, 2026) began with the free energy principle applied locally to a network of nested subsystems. The natural follow-up question is what physical conditions enable matter to function as a single unit in such networks. We have addressed this in a derivation that will be published shortly (as of v0.9.2 of this document). This shows that the minimal form of the free energy principle governing eANN units emerges directly from basic, physically plausible assumptions (Section 3 gives a high-level summary; manuscript draft is available upon request). The theory is stated at the level of the nonparametric information geometry of physically plausible dynamical systems of microstates and of their slow, coarse observables, given an appropriate separation of timescales. Under these assumptions, the familiar ingredients of Bayesian brain theories appear as consequences rather than postulates. Prediction errors emerge as the only type of change that preserves normalization of probabilities over the microstates. Sigmoid response functions arise from bounded observable ranges. The fluctuation-dissipation theorem naturally grants noise the meaning of representational precision. Consequently, perception corresponds to fast relaxation along information-geometric fibres, whereas learning is slow relaxation over the base manifold that aligns predicted with observed correlations (Figure 4). This establishes a formal, constructive path from the physical properties of matter to networks that encode priors as attractors and learn through local rules.

The programme is organized around three hypotheses, each tested in one line of work:

1. **Matter can infer and learn (Line A, "Approach and theory").** At their most basic level, inference and learning are capabilities of suitably organized matter, rather than algorithms applied to it. Line A finalizes the unit-level theory and extends it to hierarchy, action, planning, as well as thermodynamic cost and implementability of emerging hardware.
2. **Biology exploits this across scales (Line B, "Brains").** Living systems use and exploit this fundamental from of inference across various scales and, therefore, show measurable signatures of it. Line B fits eANNs to fMRI data as generative models of large-scale brain dynamics, tests the theory, and investigates which computations are handled by large-scale attractors versus processes below them, using established neuroscientific and behavioral paradigms. The current programme presents learning, extinction, expectations, and pain modulation as focal points - not only because these yield an excellent testbed for our framework but also due to their fit to the authors current collaborative network. Extensions to other indications are not only possible, but desirable.
3. **Machines can do the same, natively (Line C, "Computers").** Line C capitalizes on the emergent computational capacities of eANNs and aims to establish them as an alternative route to brain-like AI. We will evaluate eANNs on a neuroscience-inspired benchmark suite that allows drawing direct links to Line B, and explore scalability on emerging thermodynamic hardware which treat noise as a resource rather than a hindrance.

The programme does not claim a complete account of the brain's complexity, but aims for a stronger hold on its foundations: (A) predictions emerging from a single construction that may apply to both brains and machines across scales; (B) generative models of large-scale brain dynamics where each component carries a distinct computational role and a useful Bayesian interpretation; and (C) learning machines where continual learning and uncertainty are intrinsic rather than engineered or designed features. Since the three lines examine the same construction from different angles, they refine each other: the theory directs what experiments should seek, brain data reveal which forms biology employs, and machines provide controlled tests. The potential benefits are correspondingly wide-ranging. Neuroscience would gain a unifying principle that links its various scales, while AI would find a path toward brain-like systems capable of continual learning and representing their own uncertainty, potentially at a fraction of current energy costs. If inference and learning are indeed properties of suitably organized matter, then understanding the brain and engineering intelligent machines ultimately become a single scientific challenge.

1. Introduction

1.1 What brains do and machines still struggle with

Modern AI writes, reasons, generates images and delivers rigorous mathematical proofs, but its progress towards biological intelligence is strikingly uneven. Large models are trained offline, on vast data, with global loss functions and with methods shaped by the constraints of von Neumann hardware (the "hardware lottery", Hooker (2021)). At least four capacities that brains show routinely remain hard for them.

The first is *continual (or test-time) learning*. In brains, learning and inference are inseparable and never stop, whereas artificial networks trained on new data tend to overwrite what they knew (catastrophic forgetting). The same limitation shows up within a single long task: in a recent comparison, language-model agents given a day of compute fell behind the gains that repeated sampling would predict, whereas expert humans kept improving at least over two weeks (Mang et al., 2026), and probably for far longer. From the perspective developed below, this is less a matter of protecting old weights than of missing timescales: inference runs, but there is no slower level of structure that keeps adapting while the task is performed, and no level slower still that consolidates what was learned (Sections 3, 4.3 and 5). The second is *uncertainty*: brains act on calibrated uncertainty (Knill and Pouget, 2004; Pouget et al., 2016), while most artificial networks represent it poorly (Guo et al., 2017) or only through add-on machinery (Gal and Ghahramani, 2015; Lakshminarayanan et al., 2016). The third is *frugality*. Brains learn with local plasticity driven by noisy signals, on a tiny energy budget (Attwell and Laughlin, 2001), whereas backpropagation needs globally orchestrated error transport (Lillicrap et al., 2020) and runs on deterministic hardware that spends much of its energy suppressing noise (Strubell et al., 2019; Hooker, 2021). The fourth is *genuine agency*: brains plan and act on an internal model of the world. In current AI, despite impressive progress, agency and active, goal-directed behaviour may still be largely mimicry (Shanahan et al., 2023; Pezzulo et al., 2024), or behaviour engineered on top of a pretrained model by reinforcement learning (Ouyang et al., 2022), rather than something that arises from the model itself (Kambhampati et al., 2024).

Brains show that all of this is physically achievable, and neuroscience has identified many of the mechanisms involved, from local plasticity (Bi and Poo, 1998) and replay (Wilson and McNaughton, 1994) to attractor dynamics (Khona and Fiete, 2022). What AI lacks is therefore not an existence proof but a principled foundation: an account of why these mechanisms work that is general enough to carry over to other substrates.

1.2 Lost across scales

Few objects in science have been described at as many scales as the brain: molecules and synapses, neurons and circuits, regions and whole-brain networks, each with its own methods and its own models. What is still missing is an agreed and testable account, at the level of physics, of how living matter can infer and learn at all, and why similar capacities recur at very different scales, from single cells to whole brains (Lyon, 2005; Levin, 2019).

One symptom is that models at different scales rarely connect. Bottom-up approaches build whole-brain models from local biophysics, with neural masses, neural fields or mean-field approximations. They are mechanistic, but every modelling choice multiplies the number of plausible variants, and macro-scale data alone can rarely decide between them. Top-down approaches describe macro-scale phenomena, such as functional connectivity and resting-state networks, without a mechanism that generates them. The two research cultures rarely meet (Rosas et al., 2025). A principle that holds, or is at least approximated, at every scale would give both a common reference, against which bottom-up mechanisms and top-down descriptions could be tested.

1.3 Why existing frameworks do not close the gap on their own

Several frameworks supply parts of such a principle. None, we argue, supplies all of it, and the approach proposed here is meant to complement them rather than replace them (Section 10 compares them in more detail).

- **Biophysical whole-brain models** are mechanistic, interpretable and the most direct link between large-scale dynamics and physiology (Deco et al., 2008; Sanz Leon et al., 2013; Breakspear, 2017). But they are not designed to learn or to perform the functions whose neural basis they describe,

which limits what they can offer AI, and they take the local mechanism as given at one privileged scale. We aim to add a scale-agnostic perspective (Section 4.2).

- **Task-optimized deep networks ("neuroconnectionism")** have given neuroscience its most successful models of sensory representations, and a productive way of comparing brains and machines on the same tasks (Richards et al., 2019; Doerig et al., 2023). Their strength is task fidelity. Their neurobiological validity is partial, however (learning by backpropagation, idealized units, rarely generative models of whole-brain dynamics), and architectures, objectives and learning rules are chosen heuristically and by fit to data rather than derived from principles. They show which functions a brain-like system can learn, but not why suitably organized matter comes to infer and learn in the first place.
- **Energy-based associative memories**, from Hopfield networks and Boltzmann machines (Hopfield, 1982; Ackley et al., 1985) to dense associative memories with very large capacity and a close link to attention in transformers (Krotov and Hopfield, 2016, 2020; Ramsauer et al., 2020), show how far memory and computation can go with local interactions. Their energy function and learning rule are chosen by the designer, typically for capacity, and few are tied to brain data. The networks proposed here recover them as special cases for binary units, symmetric couplings and equilibrium (Section 3.4), but derive their energy function and learning rule instead of choosing them.
- **Physical learning and neuromorphic computing** move learning into the substrate itself: equilibrium propagation and coupled learning (Scellier and Bengio, 2017; Stern et al., 2021), classical and neural cellular automata (Wolfram, 1984; Mordvintsev et al., 2020), and spiking networks (Maass, 1997) with rules such as spike-timing-dependent plasticity (Bi and Poo, 1998), which promise brain-like energy efficiency on neuromorphic hardware (Furber et al., 2014; Davies et al., 2018; Roy et al., 2019; Marković et al., 2020). Their learning rules, however, are mostly designed by hand or approximate backpropagation (Neftci et al., 2019; Bellec et al., 2020). Work that derives spike-based plasticity from likelihood or expectation maximization (Pfister et al., 2006; Nessler et al., 2013) and reads spikes as samples from a posterior (Buesing et al., 2011) comes closest to the inferential reading adopted here (Section 4.4), and even the circuit for gradient-based credit assignment can self-assemble under uniform local plasticity (Liao et al., 2026). In all of these, however, the objective or the plasticity rule is postulated rather than derived from the physics of the unit.
- **Large-scale attractor and energy-landscape models of brain dynamics**, in a tradition that goes back to Amit (1989), include multistable and ghost-attractor regimes of connectome-based mean-field models (Deco and Jirsa, 2012; Golos et al., 2015), coupled-oscillator models (Cabral et al., 2011), maximum-entropy (Ising-type) landscapes fitted to fMRI (Watanabe et al., 2013; Ezaki et al., 2017; Gu et al., 2018), Hopfield-type networks built from mouse cortical connectivity (Fasoli et al., 2026) or estimated from MEG (Gosti et al., 2024), attractor-state perspectives on the default mode network (Poerio and Karapanagiotidis, 2025), and our own connectome-based attractor networks of the human brain (Englert et al., 2026), now also applied by independent groups (Watters et al., 2026). They capture real features of brain dynamics, but most are phenomenological: couplings are taken from connectivity or fitted to data, and dynamics and storage rules are chosen rather than derived. This holds even where the landscape is fitted directly to regional time series. A recent study relates the geometry of such a landscape to attention (Song et al., 2026), but the form of its dynamical model is again chosen, so the attractors depend on modelling and fitting choices that the data alone constrain poorly. Why brain dynamics take this form, what the attractors compute, and how such landscapes could serve AI as trainable learners remain largely open.
- **The free energy principle (FEP) and active inference** offer the broadest first-principles account of self-organization as inference (Friston, 2010; Friston et al., 2023a). Their route is top-down: from a whole system that persists, separated from its environment by a Markov blanket, to internal states that can be read as encoding beliefs about external ones. As a principle, the FEP is tested through the process theories built on it, such as predictive coding and active inference, and recent work has specified the conditions under which its inferential reading holds (Biehl et al., 2021; Aguilera et al., 2022; Bruineberg et al., 2021). The present programme approaches it from the bottom up. From a handful of explicit physical assumptions about a single unit (Section 3), it derives, we think plausibly, the minimal form of the FEP that elementary units of inference and learning require (manuscript in preparation), and in this setting the principle and its Bayesian readings hold rigorously. Everything beyond this minimal form, from structured priors to sequence learning, comes from composing such units into attractor networks (Section 4), as described by Spisak and Friston (2026). The FEP is thus a guiding principle, an interim result derived at the level of the unit, and

a link to Bayesian brain theories and active inference. What the construction adds is a concrete theory with falsifiable predictions.

1.4 What is missing: a substrate-independent account of inference and learning across scales

Taken together, Sections 1.1 to 1.3 point to one gap. What is missing is not another model of a particular phenomenon at a particular scale, but a substrate-independent, that is, geometric, account of how collective dynamics with separated timescales organize into increasingly complex forms of inference, learning and action. Such an account would describe a physical candidate phenomenon for cognition: one that evolution did not have to invent, only to exploit (Fields and Levin (2020); Levin (2023); see also Fields et al. (2021), Fields and Levin (2022)), and one that may offer guidance towards brain-like artificial cognition. Our approach breaks this challenge down into two parts: minimal units, and their composition (Figure 1).

The first part is the elementary building block, the account that Section 1.2 found missing: how a piece of matter can infer and learn at all. Rather than postulating inference, or establishing it for whole systems under conditions that are hard to verify (Section 1.3), it calls for a derivation from assumptions about a single unit that can be checked for a given substrate, and that can fail. It also has to be stated in terms that do not depend on any particular substrate. We use the information geometry of a system's slow, coarse observables, in which the same adaptive, self-aligning structure can be recognized in different substrates, implementations and scales: in molecular networks and synapses, in neurons and circuits, in brain regions, and in engineered hardware (Section 3).

The second part is composition: how such units, coupled together, give rise to new computational capacities at the next spatial and temporal scale, and whether the collective is again a unit of the same kind. This is where the gaps of Section 1.1 belong. Continual learning, calibrated uncertainty, frugal local learning and agency are not properties of single units. If they arise from general units, they must arise from the collective dynamics: from the structure the units build together, and from the slower timescales on which that structure changes. Networks built in this way already show several of these properties, among them priors stored as attractors, resistance to catastrophic forgetting and sequence learning (Section 2). What has been missing is a formal account of why composition produces them, how far it carries across scales, and which of the capacities of Section 1.1 it can deliver (Sections 4 and 5).

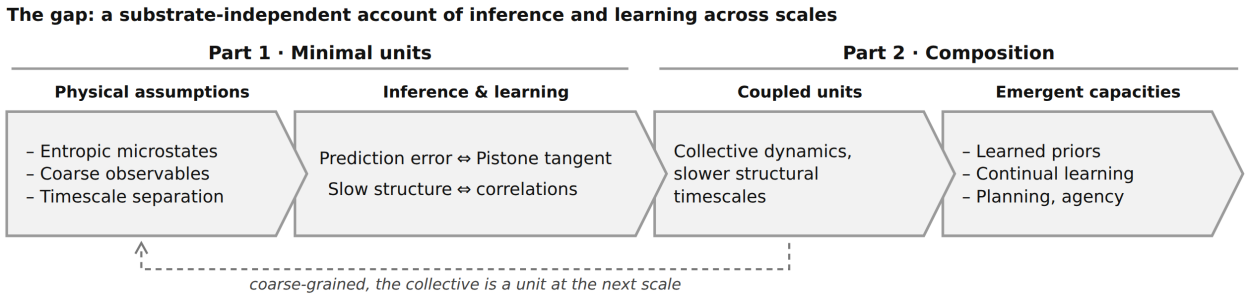


Figure 1: ***The gap, and how we break it down.*** *What is missing is a substrate-independent account of inference and learning across scales. We split it into two parts.* ***Part 1*** *(minimal units) starts from three physical assumptions about a single unit: entropic microstate dynamics, seen through coarse observables, with separated timescales. From these follow inference and learning: prediction errors are tangent vectors of the Pistone manifold, and the slow structural coordinates encode correlations (Section 3).* ***Part 2*** *(composition) goes from coupled units, with collective dynamics and slower structural timescales, to the capacities of Section 1.1 (Sections 4 and 5). Coarse-grained, the collective is again a unit, so the construction repeats at the next scale (dashed arrow).*

Both parts also have to build on the strengths of the frameworks reviewed in Section 1.3, which the new account is meant to complement and connect rather than replace. Its models should work as learners, as task-optimized networks do. They should be fitted to brain data, as biophysical and landscape models are. Their rules should follow from principles rather than from choice, as the FEP demands. And they should be implementable in physical substrates, as physical and neuromorphic learning aims to be. Each of these frameworks already delivers on its own requirement; what a common construction can add is a way to meet them together.

This is a programme about roots, not about the whole tree. The brain's many cell types, its molecular machinery, its developmental history and its individual variability will remain a challenge for a long time.

What we aim for is a much better understanding of the roots from which this complexity grows, and which each biological implementation realizes in its own way. The rest of this document describes how far our approach has taken both parts, and what remains to be done.

1.5 How to read this document

Two routes through the same material. The argument has a conceptual order and a chronological one, and this document uses both. Conceptually, it runs from the unit to the network: a single unit infers and learns under stated physical assumptions, such units compose into networks, and these networks are then fitted to brains and built into machines. This is the order of the executive summary. Chronologically, the networks came first (Spisak and Friston, 2026). They were derived from the free energy principle, simulated and fitted to human brain data, and the unit-level construction was developed afterwards to ground them in physics, which turned variational free energy minimization from an assumption into a derived result. The main text follows this chronological order. It starts from what is published (Section 2), then turns to the newer foundation underneath it (Section 3, an outline of unpublished results), and the two routes meet in Section 4, where the same networks are obtained again, this time from below (a recap of Spisak & Friston, 2026).

After these three sections, Section 5 asks what attractors are, and Section 6 collects biology-like phenomena that may follow. Section 7 lists the predictions with their failure criteria, Section 8 the technical plan, and Sections 9 and 10 the risks and the relation to other approaches. Section 11 collects further-reaching ideas on which nothing else depends. How the document came about, and how to comment on it or contribute, is described in *About this document: its origins, status and how to contribute.*

To keep the status of each statement visible, key claims carry a tag:

Tag	Meaning
PUBLISHED	Shown in peer-reviewed work, mainly Spisak and Friston (2026) and Englert et al. (2026), as of v0.9.2 of this document
IN PREPARATION	Derived or simulated in work being prepared for publication (draft available upon request)
HYPOTHESIS	A heuristic expectation, partly supported by simulations or data, that the programme will formalize and test
SPECULATIVE	An appealing but distant possibility, flagged so that it is not mistaken for a claim

2. Where we stand: what is already established

The programme does not start from zero. Two peer-reviewed papers form its published core, one theoretical and one empirical, and they show that the networks at the centre of this white paper (Figure 2) exist, work and can be fitted to human brain data. In the logical order of the argument these networks are the end point; historically, they came first (Section 1.5).

In theory and simulation PUBLISHED. Spisak and Friston (2026) derived a class of stochastic attractor networks by applying the FEP recursively to a network of nested subsystems, without hand-designing either the inference or the learning rule (Figure 2). The FEP was an assumption at this stage; Section 3 derives its minimal form from the physics of a single unit. Each node infers by combining its own bias with weighted evidence from the other nodes, and each coupling learns by contrasting observed with predicted correlations, a local, predictive-coding rule. The paper showed, analytically and in simulations, that the attractors of such networks act as Bayesian priors (Figure 2). Additionally, we found that the learning rule does something that was never put into it: even when the inputs are strongly correlated, it drives the attractors towards approximate orthogonality (Figure 2). The networks find a decorrelated, near-orthogonal code on their own, the kind of representation that efficient-coding theories have long proposed for the brain, and that machine learning usually has to enforce with additional objectives or constraints. Orthogonal attractors interfere least with one another, which fits two further findings: the networks generalize to new inputs, and they resist catastrophic forgetting. Sequential input, finally, produces asymmetric couplings and directed sequence dynamics (Figure 2). The simulation code is available at: https://github.com/pni-lab/fep-attractor-network.

In the human brain PUBLISHED. Englert et al. (2026) built networks of this class directly from human functional connectivity. Their attractors were approximately orthogonal, held up against strict null models, and

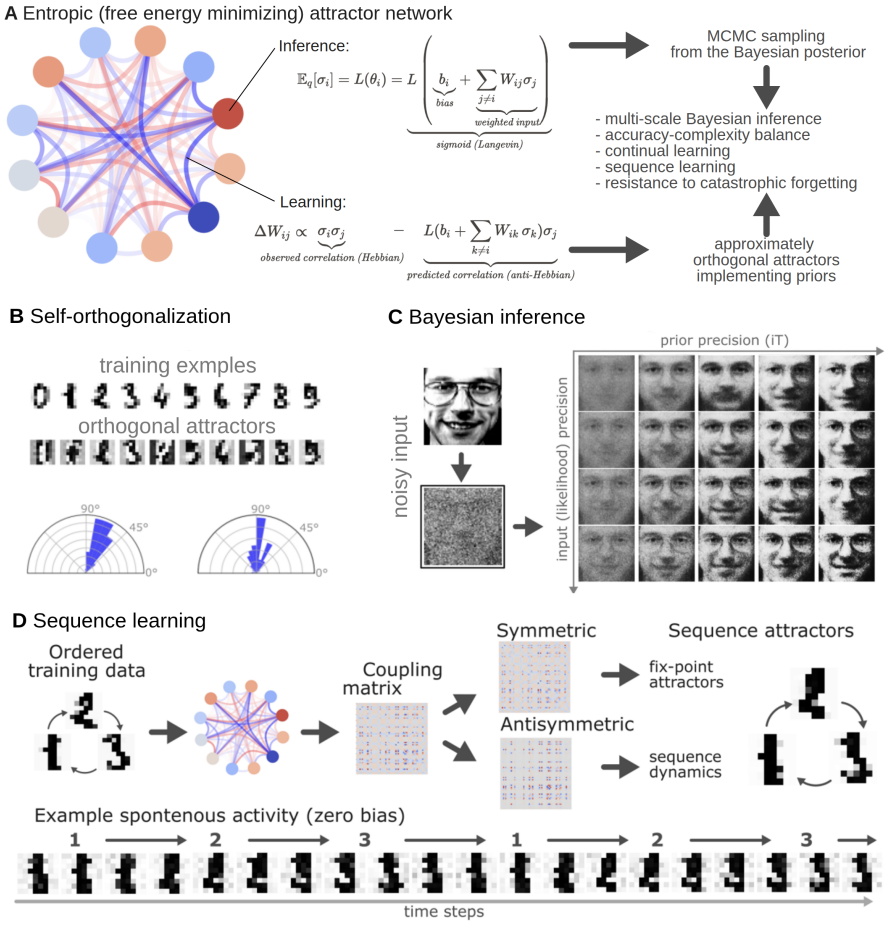


Figure 2: ***Entropic attractor networks: two local rules, derived rather than chosen, and what they produce.*** ***A*** *A network of stochastic units with bounded states σ_i, connected by couplings W_{ij}, and the two rules that define it. In inference, each unit passes its bias b_i, which acts as a prior, plus the coupling-weighted input from the other units through a sigmoid response function L. This is the Langevin function for a uniform (or, without loss of generality, uniformized) reference measure, and it reduces to tanh for binary units. In learning, each coupling changes by the difference between the correlation it observes (Hebbian term) and the correlation the network predicts (anti-Hebbian term), a local, predictive-coding rule. Neither the rules nor the response function are designed by hand. They were derived from the free energy principle applied to a network of nested subsystems (Spisak & Friston, 2026), and they follow from elementary physical assumptions about a single unit (Section 3). The noisy dynamics sample from the Bayesian posterior, and learning drives the attractors towards approximate orthogonality, so that they act as priors. For binary units, symmetric couplings and equilibrium, the network reduces to a stochastic Hopfield network or Boltzmann machine (Section 3.4).* ***B*** *Self-orthogonalization. Trained on correlated examples (handwritten digits), the network stores attractors that are approximately orthogonal: the angles between pairs of attractors (right) concentrate near 90ř, more tightly than those between the training examples (left).* ***C*** *Bayesian inference. Given a noisy version of a learned pattern, the network's state combines prior and evidence: what it settles on depends on the precision of the prior (inverse temperature, iT; left to right) and on the precision of the input, or likelihood (top to bottom).* ***D*** *Sequence learning. Trained on an ordered sequence (1, 2, 3), the network learns a coupling matrix whose symmetric part encodes fixed-point attractors and whose antisymmetric part encodes the transitions between them. Without input (zero bias), its spontaneous activity cycles through the learned sequence. Simulations in B–D from Spisak & Friston (2026).*

provided initial evidence that the landscape corresponds to canonical large-scale brain networks, replicates across datasets, and captures task-related changes and altered dynamics in clinical conditions. Applying the networks at the scale of brain regions relies on the scale-agnostic nature of the framework: the construction is expected to keep its form under coarse-graining (Section 4.2), so that brain regions, and the slow signals that fMRI resolves in space and time, can be modelled as units of the same family. This property is currently a well-motivated heuristic, and formalizing it is a goal of Line A. The networks have also begun to be used outside our group: in a fully independent study, Watters et al. (2026) applied them, with the pretrained embedding of Englert et al. (2026), to resting-state fMRI of individuals with high smartphone addiction scores. The attractor states supplied the reference frame (internal versus external context) around which the study's clearest, multiple-comparison-corrected group differences emerged: altered transitions between the default mode and attention networks. Beyond our framework, a connectome-based Hopfield-type model of mouse cortex predicts fMRI co-activation patterns from its attractor landscape (Fasoli et al., 2026), and a recurrent Hopfield network with asymmetric couplings, fitted to MEG, estimates meso-scale effective connectivity and predicts brain dynamics (Gosti et al., 2024). Our open-source package *attractome*, the successor of the now deprecated connattractor package used by Watters et al. (2026), makes such fits available to others.

From connectivity to time series (IN PREPARATION). Englert et al. (2026) initialized the couplings from the negative inverse covariance (the precision matrix) of the regional time series. We have since developed a more principled approach, implemented in attractome: the networks are fitted to the regional time series directly, by score matching (Hyvärinen, 2005). Besides the couplings, the fit also estimates each region's bias, its own, possibly time-varying input. This separates two levels of activity in the same data: network-level dynamics, carried by the couplings and their attractor landscape, and region-intrinsic or upstream activity, carried by the biases (graphical abstract, Line B; Section 5.2). Because the biases can be modelled with arbitrary design matrices, the separation also holds for complex task designs. The approach also yields reliable attractor states at the level of individuals. Figure 3 gives a preview of these newer, not yet published analyses (see also Section 8.2).

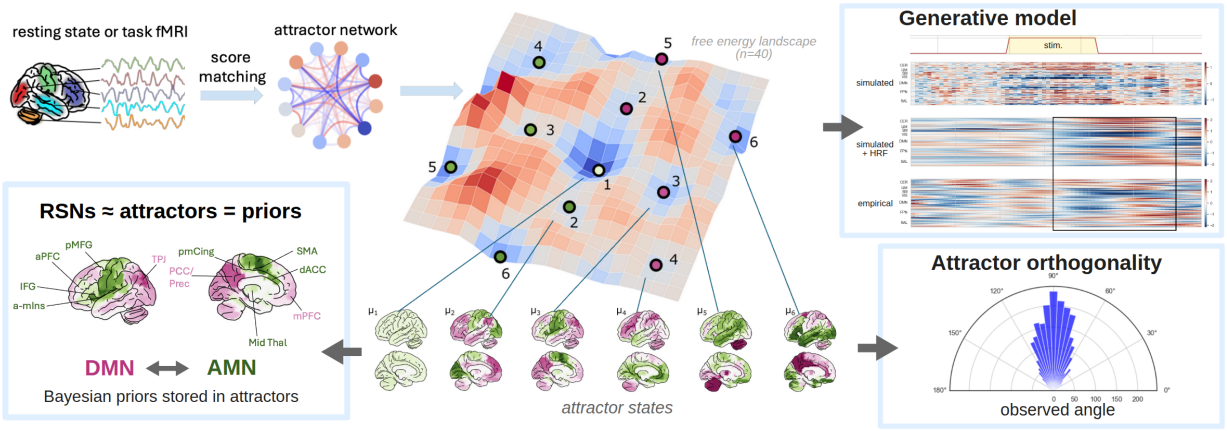


Figure 3: ***Entropic attractor networks fitted to human fMRI.*** *Top left:* networks of the same family as in Figure 2 are fitted to resting-state or task fMRI time series of brain regions by score matching. *Centre:* the fitted network defines a free-energy landscape whose minima are attractor states (μ_1–μ_6), shown on the cortical surface below it. *Bottom left:* the attractors resemble canonical resting-state networks, such as the default mode network (DMN) and the action mode network (AMN), and in the framework they are the Bayesian priors the network has learned. *Top right:* as a generative model, the network reproduces regional brain activity: simulated dynamics, convolved with a haemodynamic response function, resemble the empirical time series, here around a stimulus block. *Bottom right:* the fitted attractors are approximately orthogonal, with pairwise angles concentrated around 90ř, as the learning rule predicts (Figure 2). The model is applied here at the spatial scale of brain regions and the temporal scale of the BOLD signal. That it applies at this scale at all rests on the scale-agnostic nature of the framework: the construction is expected to keep its form under coarse-graining (Section 4.2), a property that is currently heuristic and will be formalized in Line A. Preview of unpublished analyses (IN PREPARATION): the data are those of Englert et al. (2026), but the network is fitted with the attractome package, by score matching to the regional time series rather than derived from functional connectivity as in the original paper. The resulting couplings are highly similar to the published ones.

What comes next. The derivation of Spisak & Friston (2026) shows that if a system can be described as nested subsystems under the FEP, these inference and learning rules follow. Which physical conditions a

piece of matter has to meet to behave as one of the nodes is the subject of Section 3. On this basis, Line A extends the theory to hierarchy and action, Line B tests the predictions that go beyond orthogonality, and Line C brings the networks to physical hardware.

3. A minimal inferring and learning unit

Section 2 summarized our results on self-orthogonalizing attractor neural networks, as published in Spisak & Friston (2026). This section goes one level down, to the single unit from which they can be built: it states the construction of the minimal unit and what follows from it. The aim is a sufficient route from physics to inference and learning, not a necessary one: a minimal and physically plausible model of how matter can be organized to infer and learn. The construction also changes the status of the free energy principle. In Section 2, the FEP was the starting point from which the networks were derived. Here, its minimal form, a single unit that minimizes variational free energy, follows from the unit's physics. The FEP thus turns from an assertion into an important interim result, and the networks of Section 2 come to rest on stated physical assumptions rather than on the principle itself (Section 4). The full derivations are in the manuscript "From physical dynamics to perception and learning" (Spisak et al., in preparation; available on request) IN PREPARATION. Here we give the assumptions, the results in informal form, and the known cases they reduce to.

3.1 Why this is possible now

That inference might be a property of matter, and not only of minds, is an old intuition. Several developments, some long established and some recent, now fit together and make it possible to turn it into a construction.

A geometry in which prediction errors are natural objects. The non-parametric exponential manifold of Pistone and Sempi (1995) describes the space of all normalized probability densities of a system, without fixing a parametric model in advance. On this manifold every tangent vector is centred: it has the form "actual minus expected". Specifically, writing a density in exponential coordinates, $p = e^{u-\psi(u)}\,p_0$ with normalizer $\psi(u) = \log \mathbb{E}_{p_0}[e^u]$, normalization alone gives

$$\partial \log p \;=\; \partial u - \partial\psi \;=\; \partial u - \mathbb{E}_p[\partial u], \qquad \text{since} \qquad \mathbb{E}_p[\partial \log p] = \int \partial p = \partial \int p = 0. \tag{1}$$

Prediction errors, which Bayesian brain theories postulate, can therefore be read as the generic way a normalized ensemble changes.

Learning as constrained maximum entropy. Jaynes' maximum-entropy principle (Jaynes, 1957) and the duality theory of exponential families (Amari, 2016) relate each parameter of a distribution to the expectation it controls. Recent work has made the correspondence between constrained maximum entropy and the FEP explicit (Ramstead et al., 2023; Sakthivadivel, 2022). This is what allows slow structural coordinates, such as couplings, to be treated as thermodynamic conjugates of correlations.

Timescale separation as a tool. Averaging and adiabatic elimination (Pavliotis and Stuart, 2008; Haken, 1983) turn a separation of timescales into a precise statement about which variables can be treated as fixed, which as equilibrated, and which as slowly drifting. In the construction this separates perception from learning.

A sharper question. The critical debate about the FEP (Biehl et al., 2021; Aguilera et al., 2022; Bruineberg et al., 2021) has clarified what a physical account of inference must specify. A construction whose assumptions are local and explicit is a well-posed contribution to that debate, rather than a rival grand theory.

A known target. Finally, the network rules of Spisak and Friston (2026) are published and a construction from below has to recover them, including the exact form of the learning rule and its timing. Until recently, there was no comparably concrete target against which a unit-level construction could be checked.

3.2 Assumptions

The construction (Figure 4) uses a small set of explicit assumptions, chosen by Occam's razor: whenever several structures could support a result, we take the least elaborate one.

1. **A normalized reference ensemble (A1).** The stationary densities of the system are positive and regular enough to admit local exponential coordinates, and environmental forcing acts as an exponential tilt of a reference state.
2. **Coarse observables with a double separation of timescales (A2).** Microscopic variables relax fastest, retained mesoscopic coordinates next, and slow structural coordinates (couplings) much more slowly. The environment varies on an intermediate timescale.
3. **Bounded coordinates and a point of view (A3).** Retained coordinates are bounded, as in any finite physical system. One of them, the exposed coordinate, is followed from its own point of view, and the others form its boundary. No Markov blanket is required: shielding from the environment is a special case, not a precondition.
4. **Minimal interaction and least-structured response (A4).** Near a generic operating point, only the leading interaction between boundary and exposed coordinate is kept, and the resting law of the exposed coordinate is the maximum-entropy law on its range.
5. **Causal renewal, for networks (A5).** Coupled units renew together, each forming its prediction from the preceding state of its boundary.

Each assumption is an empirical question for a given substrate, and each can fail.

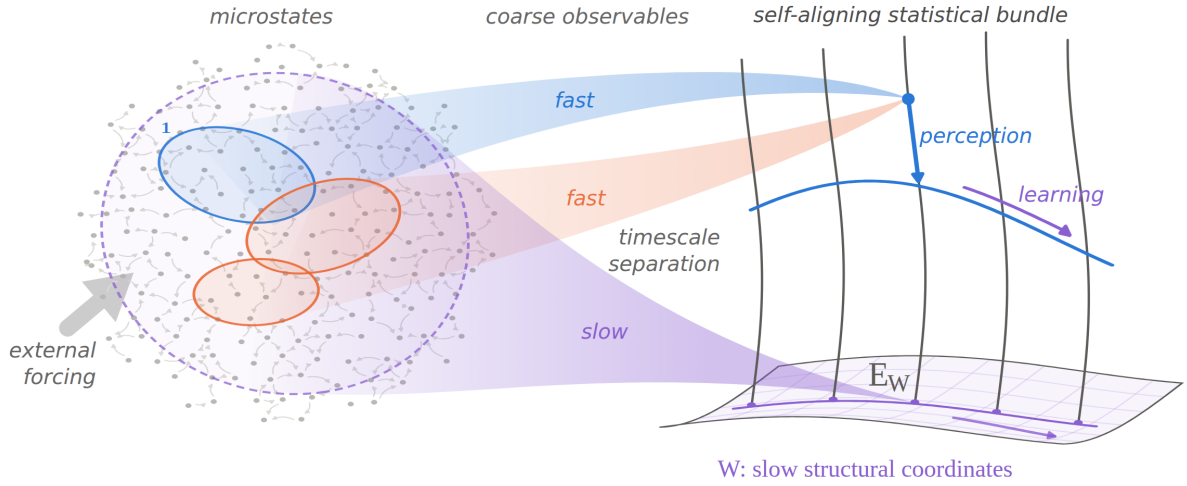


Figure 4: ***The minimal inferring and learning unit. Left*** *A physical system of many microstates (grey), driven by external forcing that tilts its reference ensemble (A1). A few bounded, coarse observables are retained (A2, A3): the exposed coordinate (blue, 1), followed from its own point of view, and the coordinates that form its boundary (orange). These relax fast. The structural coordinates W, such as couplings, shape the whole ensemble and change slowly. This separation of timescales (A2) is what separates perception from learning.* ***Right*** *The same construction in information-geometric terms, as a statistical bundle. The base manifold E_W collects the slow structural configurations, and above each of them a fibre holds the distributions of the fast coordinates. Perception is fast relaxation along a fibre towards the posterior (R4). Learning is slow drift along the base, which moves the structure until predicted and observed correlations agree (R5). Because the structure thereby aligns itself with the statistics of its environment (Section 3.4), we call the bundle self-aligning.*

3.3 Results

The results are stated informally; precise statements, conditions and proofs are given in the manuscript in preparation. For each result we name the assumptions it uses and, in one line, the route by which it is obtained.

- **R1. Bayesian structure from normalization** (A1, A2). Normalizing the tilted ensemble over fast and slow coordinates turns the reference state into a prior, the tilt into a likelihood and the normalizer into model evidence. No beliefs are assigned to microstates. *Route:* the normalization of an exponentially tilted density, read at fixed slow coordinates.

- **R2. Prediction error as a tangent** (A1–A3). Along the fast direction, the centred tangent of the exposed coordinate is its prediction error, $\epsilon = \mu$ - E[μ | s, W]. It follows from preserving normalization alone. *Route:* the tangent space of the non-parametric exponential manifold.
- **R3. The response function from boundedness** (A1, A3, A4). The unit's mean response is the derivative of the log-partition function of the bounded maximum-entropy family: the Langevin function L(x) = coth x - 1/x, which reduces to tanh for a binary coordinate. No activation function is chosen by hand. *Route:* Legendre duality of the bounded maximum-entropy family.
- **R4. Perception as fast relaxation** (A1–A4). The fast relaxation of the mesoscopic coordinates moves the distribution of the unit's state towards the posterior; it is descent on variational free energy at fixed structure. *Route:* adiabatic elimination of the microscopic variables.
- **R5. Learning as slow relaxation** (A1–A4; timing from A5). Timescale separation makes the expected correlations of the mesoscopic coordinates the Legendre duals of the slow structural coordinates. The structure then climbs the average evidence, driven by a contrastive learning signal: observed minus expected correlation, a Hebbian minus an anti-Hebbian term. *Route:* a dynamical form of Jaynes' constrained maximum entropy, by averaging over the fast timescale.
- **R6. Uncertainty from fluctuations** (A1–A4). At both timescales, fluctuations sample the corresponding posterior. For the fast state, variance equals gain, a fluctuation–dissipation relation; for the slow structure, this is the idea of synaptic sampling (Kappel et al., 2015). *Route:* the second derivative of the log-partition function, which is both a variance and a susceptibility.

The two centred quantities that drive perception and learning can be written compactly:

The construction in two sentences
Perception. A prediction error, geometrically, is nothing more than a normalization-preserving, and therefore expectation-centred, tangent vector on a non-parametric statistical manifold.

$$\varepsilon_\mu = \underbrace{\mu}_{\text{actual state}} - \underbrace{\mathbb{E}[\mu \mid s, W]}_{\text{expected state (prediction)}} \tag{2}$$

Learning. A synapse is the thermodynamic Legendre conjugate of a correlation.

$$\psi(W) = \sup_q \left\{ \underbrace{W}_{\text{structural parameter}} \underbrace{\mathbb{E}_q[s\,\mu]}_{\text{expected correlation}} - \underbrace{D_{\mathrm{KL}}(q \,\|\, q_0)}_{\text{relative entropy}} \right\} \tag{3}$$

$$\varepsilon_W = \underbrace{s\,\mu}_{\text{observed correlation}} - \underbrace{\mathbb{E}[s\,\mu \mid W]}_{\text{expected correlation}} \tag{4}$$

Here ψ is the log-partition function (a free energy) and q_0 the reference law of the unconstrained dynamics. The structural parameter and the expected correlation form a Legendre pair, with relative entropy as the dual potential. Since $\partial\psi/\partial W = E[s\mu \mid W]$, the learning error is the gradient of the dual objective $W{\cdot}s\mu$ - $\psi(W)$: learning climbs it until expected and observed correlations agree, which is Jaynes' constrained maximum entropy in dynamical form. Synapse is meant in the general sense: any slow coupling between units.

3.4 Known cases, and what the unit does not do

A new construction should reduce to known results where those apply. For binary units, symmetric couplings and equilibrium, R3 gives the familiar tanh response of Ising-type models, and R5 gives a contrastive, Hebbian minus anti-Hebbian rule of the same form as Boltzmann-machine learning (Ackley et al., 1985). For a continuous bounded coordinate, R3 gives the Langevin function of the classical paramagnet. R6 is the fluctuation–dissipation relation of the exponential family (Kubo, 1966). What is new is not these ingredients but their joint derivation from one set of physical assumptions, their Bayesian reading, and the timing of the learning signal. In the construction this timing follows from causal renewal (A5), and in the networks of Spisak and Friston (2026) it is predicted to be what makes the attractors orthogonalize (P3).

Just as important is what the construction does not claim. It shows one route from physics to inference and learning, not the only one, and it does not show that brains, or any particular material, take this route. Where its assumptions fail, for example without a clear separation of timescales, the inferential reading is not guaranteed, and the learning signal becomes biased.

The construction also defines a *family* of units and networks, not one set of equations. Three ingredients come from the substrate and select a member of the family: the reference measure of the microstates, the bounds of the retained coordinates (which fix the response function, the Langevin function in general and tanh only for a binary coordinate), and the type of intrinsic interaction between mesoscopic coordinates. What all members share is the Bayesian structure, the centred prediction error, the Legendre-dual learning signal and the signatures that follow from them. When this white paper speaks of the same rules at different scales or in different substrates, it means rules of this family.

Nor does the construction require finely tuned structure. Structural coordinates may start far from any useful configuration; the slow dynamics then move them until expected and observed correlations agree, so that the structure aligns itself with the statistics it is exposed to. How far this self-alignment can go is limited by the substrate, by the range, precision and timescales of the structural coordinates it offers. A physical system may thus obey the construction only weakly, with little headroom, and evolution and development can be read as processes that build substrates with more headroom (Sections 1.4 and 11) HYPOTHESIS.

Why "entropic". The name refers to the dynamics, not only to their endpoint. A long tradition treats relative entropy as what drives motion, not just as a description of where a system ends up: fluxes driven by entropy gradients in irreversible thermodynamics (Onsager, 1931), the Fokker–Planck equation as a gradient flow of relative entropy (Jordan et al., 1998), and the split of dynamics into a reversible, energy-driven part and an irreversible, entropy-driven part (Grmela and Öttinger, 1997), which mirrors the decomposition used in Section 4.5. In this sense, inference in the units is a relaxation that lowers relative entropy towards the posterior, learning is constrained maximum entropy, and the solenoidal part of the dynamics leaves relative entropy unchanged. As with entropic forces, what drives all three is the multiplicity of hidden microstates, not a designed interaction. The programme of Entropic Dynamics (Caticha, 2011) shares this intuition but not the construction.

Setting, notation and the assumption–result map

Setting. A physical system has microscopic variables that are not tracked individually. A small number of coarse observables over them are retained: bounded mesoscopic coordinates, one of which, the exposed coordinate μ, is followed from its own point of view, while the others form its boundary s. Slow structural coordinates W (couplings) parametrize the interaction between boundary and exposed coordinate. The environment acts on the system as an exponential tilt of a reference ensemble q_0.

Timescales. The construction uses the ordering

$$\tau_{\text{micro}} \ll \tau_{\mu} \ll \tau_{\text{env}} \ll \tau_{W} \tag{5}$$

The first inequality allows the microscopic variables to be eliminated (perception, R4). The middle one means that the environment changes slowly enough for the state to track it, but fast enough for the

structure to average over it. The last allows the structure to be averaged over the fast and intermediate timescales (learning, R5).

Symbol	Meaning
μ	Exposed mesoscopic coordinate (a node's state), bounded
s	Boundary coordinates (the states that the exposed coordinate interacts with)
W	Slow structural coordinates (couplings); Wij in networks (denoted Jij in Spisak & Friston, 2026)
b	Bias: the unit's own field, read as a prior
q_0	Reference law of the unconstrained dynamics
ψ	Log-partition function (a free energy)
L	Langevin function, the mean response of the bounded maximum-entropy family; tanh for a binary coordinate
ε_μ, ε_W	Prediction error of the state and learning signal of the structure
D_{KL}	Relative entropy (Kullback–Leibler divergence)

Assumption–result map. A filled cell means that the result uses the assumption.

	A1 Reference ensemble	A2 Timescales	A3 Bounds, point of view	A4 Minimal interaction	A5 Causal renewal
R1 Bayesian structure	•	•			
R2 Prediction error	•	•	•		
R3 Response function	•		•	•	
R4 Perception	•	•	•	•	
R5 Learning	•	•	•	•	timing
R6 Fluctuations	•	•	•	•	
Networks (Section 4)	•	•	•	•	•

4. From units to networks: entropic attractor networks

Section 2 described the networks as they were first derived, from the FEP. Section 3 described a single unit. This section puts the two together, and the fit is seamless: couple minimal units, and the networks of Section 2 appear, with the same inference and learning rules (Spisak and Friston, 2026), now derived from stated physical assumptions rather than from the principle. Here the historical and the logical routes through this document meet (Section 1.5).

The result is a complete first-principles path, from the physics of a single unit to networks that infer and learn, and that do considerably more. Their attractors are Bayesian priors, which the learning rule self-orthogonalizes; sequences are stored in asymmetric couplings; old knowledge is protected while new knowledge is acquired; noise carries precision; and the same structure should reappear at coarser scales. None of these properties is added by design. Each follows from the collective dynamics when composing the units. Section 4.1 shows how the unit slots into the network. The rest of the section takes the properties in turn: why the construction should hold across scales (Section 4.2), the emergent computational properties (Section 4.3), noise as information (Section 4.4), and multistability, metastability and action (Section 4.5).

4.1 How the unit fits in

Coupling minimal units needs no new ingredient. A unit's boundary, the coarse signal that drives its slow structural coordinates, now consists of the other units' states. The unit's fast inference becomes the node update, through the Langevin function L, and its slow learning becomes the rule for the couplings Wij (Jij in Spisak & Friston, 2026):

$$\mathbb{E}_q[\sigma_i] = L\Big(b_i + \sum_{j\neq i} W_{ij}\,\sigma_j\Big), \qquad \Delta W_{ij} \propto \sigma_i\,\sigma_j - L\Big(b_i + \sum_{k\neq i} W_{ik}\,\sigma_k\Big)\,\sigma_j \tag{6}$$

Each term keeps the meaning that Section 3 gave it. A node's prediction error, the difference between its state and its prediction, is the centred tangent of Section 3.1. Each coupling Wij is the thermodynamic Legendre conjugate of the correlation between nodes i and j, which it adjusts until observed and predicted correlations agree (Section 3.3). The unit construction also fixes the timing of the rule: the prediction is formed from the preceding network state and paired with the renewed one. The rule belongs to the family of contrastive, local learning rules whose relation to inference is known for predictive-coding networks (Millidge et al., 2022) and, in the symmetric equilibrium case, for Boltzmann machines (Ackley et al., 1985). The network itself is a stochastic Hopfield network generalized to non-equilibrium steady states, whose noisy dynamics sample from the global posterior.

One-dimensional units, binary questions. The minimality assumption makes every node, and therefore every attractor, one-dimensional: its continuous-Bernoulli state expresses graded evidence for or against a single event. This is why, with zero biases, the sign-inverse of every attractor is also a fixed point. An attractor and its mirror image are one prior, read as evidence for or against the same latent cause.

4.2 Across scales: one model family and the same landscape at every resolution

Both the generality of the approach and its application to fMRI rest on what survives coarse-graining: the form of the construction, and the attractor landscape it produces IN PREPARATION HYPOTHESIS for brain data.

One family of forms. Timescale separation is used twice: within the unit, where it separates perception from learning (Section 3.3), and between levels, where fast modes are enslaved by slow ones that carry the long-run dynamics (Haken, 1983; Carr, 1981; Kuramoto and Nakao, 2019; Jafarian et al., 2021). In whole-brain models, this has been proposed, and demonstrated in simulations, as the mechanism by which low-dimensional structure emerges (Runfola et al., 2025). If a coarse unit, such as a column or a brain region, exposes one macro-observable, a conjugate field acting on its transition ratios again yields an exponential family, with a response function and a susceptibility equal to its variance. The particular reference family need not survive averaging, but the architecture does: one exposed statistic, a conjugate field, fluctuation–dissipation, and a Bayesian reading when the macrostate represents an external cause. What changes with substrate and scale is which member of the family applies (Section 3.4). Single neurons, cortical columns and brain regions are therefore not expected to obey identical equations, but rules of one family, which share its signatures. Such rules may thus apply from single neurons, where the learning rule resembles spike-timing-dependent plasticity (Bi and Poo, 1998) HYPOTHESIS, to brain regions (criterion D2, Section 10). Making this rigorous for networks is a goal of Line A (A2).

The same landscape. In fMRI network analyses, a coarser brain parcellation does not create a new network. It reports the activity of one neural system in fewer coordinates, the parcel means $\mathrm{y} = \mathrm{C}\sigma$. It is a "camera", and different atlases are different cameras pointed at the same scene. The attractors $\mathrm{p}\mu$ of the fine system are whole-brain patterns. Writing each as its regional profile $\mathrm{a}\mu = \mathrm{C}\mathrm{p}\mu$ plus a within-parcel residual $\mathrm{r}\mu$, the one essential assumption is homogeneity: on the attractors that matter, units within a parcel participate alike, which is exactly what functional parcellations are designed for (Gordon et al., 2014; Eickhoff et al., 2018):

$$p_i^\mu = a^\mu_{\alpha(i)} + r_i^\mu, \qquad \|\mathbf{r}^\mu\| \ll \|\mathbf{p}^\mu\| \tag{7}$$

Because the residuals average to zero within each parcel, fine inner products split exactly into a size-weighted coarse inner product and a residual term that homogeneity makes small:

$$\langle \mathbf{p}^\mu, \mathbf{p}^\nu \rangle = \sum_\alpha |P_\alpha|\, a^\mu_\alpha\, a^\nu_\alpha + \langle \mathbf{r}^\mu, \mathbf{r}^\nu \rangle \tag{8}$$

Three consequences follow. First, attractors that differ across parcels remain visible, those that differ only within parcels merge, and nothing new should appear. In nested atlases (Urchs et al., 2019), higher resolution splits coarse attractors into descendants that average back to them. Second, approximate orthogonality at the fine scale is inherited at the coarse scale, in the size-weighted parcel metric (analyses in unweighted or z-scored coordinates therefore use size-balanced atlases). Third, only a functional camera is useful: spatially scattered random parcels average structured patterns away, shrinking them roughly with the square root of parcel size until at most a global mode remains, whereas contiguous random parcels degrade gradually, the finest attractors first. The claim is not that a network can be fitted at any scale, but that meaningful

coarse-grainings preserve the landscape and meaningless ones erase it (Meshulam et al., 2019; Friston et al., 2021; Machta et al., 2013; Transtrum et al., 2015).

Assumptions, limits and tests. Besides homogeneity, detecting the landscape with fitted networks assumes that the slow dynamics are organized by a modest number of attractors, and that the coarse network models the observed regional process. Orthogonality is a prediction (P1, P2), not a prerequisite. Nothing assumes that a parcel is a large neuron, that couplings survive block averaging, or that coarse-graining is an exact renormalization map. After haemodynamic deconvolution, fMRI is a slow camera for slow patterns (Friston et al., 2014; Heitmann and Breakspear, 2018). Averaging cannot recover distinctions it has discarded, and residual haemodynamic mixing can mimic mixtures of successive states, which simulations must rule out before such mixtures are read as prior mixing (P10). Nested atlases in population and precision fMRI (Gordon et al., 2017) let us test this directly, in five steps. First, we fit networks at a fine and at a coarse resolution, and check that the coarse attractors are the fine ones seen through the coarser atlas. Second, we check that, at finer resolution, coarse attractors split into sub-patterns rather than being replaced by new ones. Third, we measure how uniformly each attractor is expressed within parcels. Fourth, we test whether the near-orthogonality of the attractors carries over from the fine to the coarse scale, against null models of matched size. Fifth, we repeat the analysis with random parcellations, which should erase the landscape rather than preserve it. The range of resolutions is fixed in advance by test–retest reliability, and the matching is also done within single participants (B1; P18–P20).

4.3 Emergent properties

Once the units are coupled and learning, the network begins to show capacities that no single unit has. None of them is built into the units or written into the learning rule. They appear only when many units, each following the same local rules, interact. They also build on one another. These emergent properties are the bread and butter of the framework. Most of them were first described by Spisak and Friston (2026), and they are what makes the whole programme viable: if the construction is right, such properties should be visible in biological systems, and they may guide the development of more brain-like artificial ones. We collect and discuss them below (Figure 5), each tagged with its current status.

- **Attractors as priors** PUBLISHED. Attractor states store the network's priors, and ongoing activity combines them with current input into a posterior (Figures Figure 2 and Figure 5). Local inference at the nodes gives rise to inference at the level of the whole network.
- **Self-orthogonalization, the key property** PUBLISHED theory, in-silico demonstrations, first evidence in human fMRI. Even for correlated inputs, the predictive-coding rule drives attractors towards approximate orthogonality: a new pattern is stored as its residual from what is already known (Figures Figure 2 and Figure 5). The network thereby approximates a Kanter–Sompolinsky projector network (Personnaz et al., 1986; Kanter and Sompolinsky, 1987), whose attractors coincide with the leading eigenvectors of the coupling matrix, but with a local learning rule. Offline routes towards similar couplings are known as well: Hebbian unlearning (Hopfield et al., 1983) and "dreaming", which provably drives Hebbian couplings towards the projector (Fachechi et al., 2019) and can exploit correlations in the data (Serricchio et al., 2025). In effect, self-orthogonalization is an on-line, approximate principal component analysis, in the tradition of Hebbian and anti-Hebbian subspace learning (Oja, 1982; Földiák, 1990; Pehlevan and Chklovskii, 2019). Orthogonal attractors can act towards separating latent causes, much as in latent-cause models of learning (Gershman et al. (2010); applied to extinction in Section 6) and use storage capacity efficiently. They also serve as a basis for the properties discussed below.
- **Attractor–prior duality** Symmetric case: formally established equivalence; antisymmetric extension HYPOTHESIS. This is the Hopfield network–restricted Boltzmann machine equivalence (Barra et al., 2012) in disguise. Approximate orthogonality makes the network statistically equivalent to a two-layer network in which the attractors are actual hidden nodes, connected only to the original nodes (Figure 5). Our heuristic is that, with an antisymmetric coupling component, non-equilibrium steady states and solenoidal dynamics appear as within-layer connections between the attractor nodes. The appeal is direct: attractors act as integrator nodes that pool evidence from many regions and feed the result back, performing (Bayesian) information integration. A change in a single connection can then be decomposed into changes in its contribution to individual attractors.
- **Superposition and prior mixing** HYPOTHESIS observed in simulations. Linear combinations of stored priors can exist as attractors or as "ghost attractors", regions of slow dynamics without a true minimum (Deco and Jirsa, 2012). Such slow regions can do real work: in classical and dense

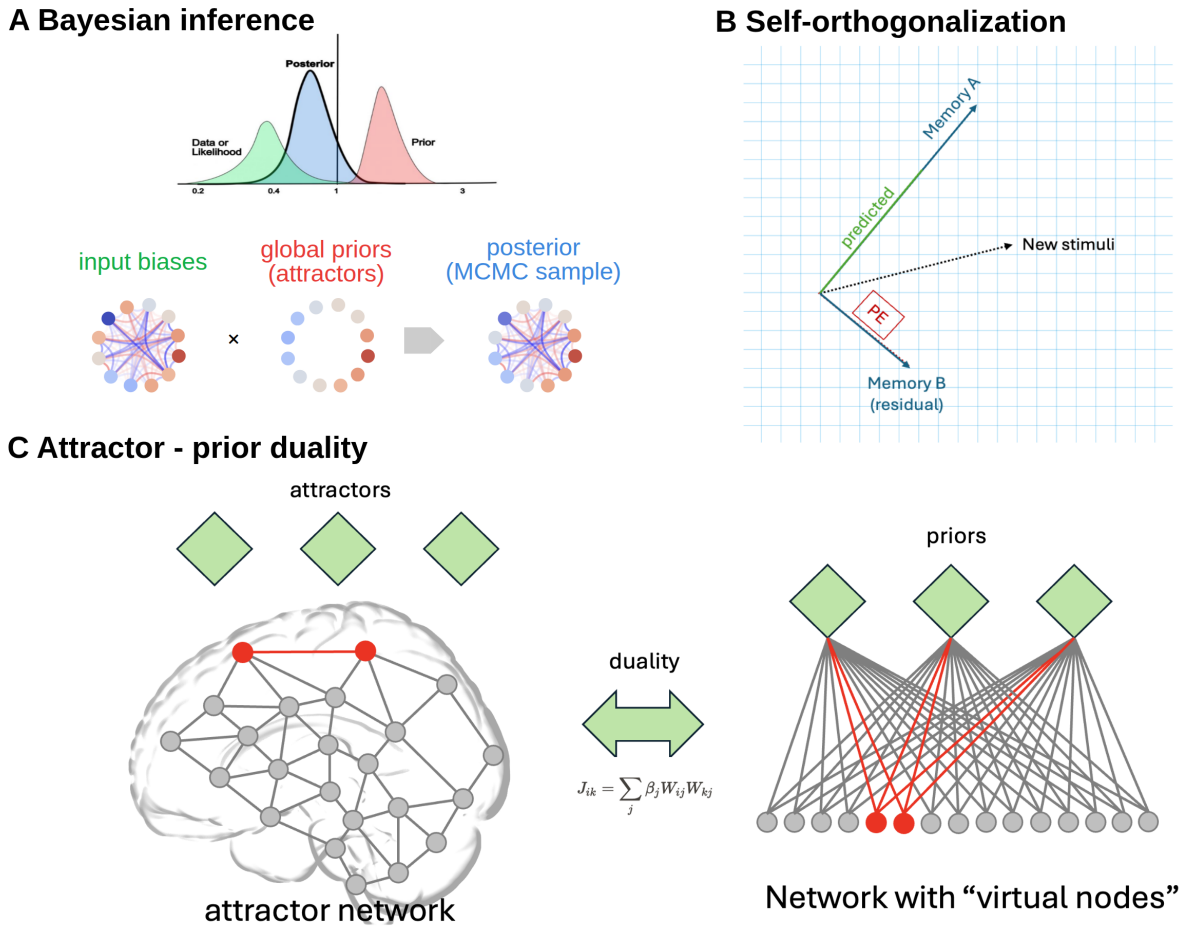


Figure 5: ***Emergent properties of entropic attractor networks.*** ***A*** *Bayesian inference. Ongoing activity combines the current input, which enters as biases, with the global priors stored as attractors. The network's stochastic states are samples from the resulting posterior (Markov chain Monte Carlo), so local inference at the nodes adds up to Bayesian inference at the scale of the whole network.* ***B*** *Self-orthogonalization. The predictive-coding rule does not store a new stimulus as it is, but as its prediction error (PE): the residual relative to what the existing memory (A) already predicts. The new memory (B) is therefore approximately orthogonal to the old one.* ***C*** *Attractor–prior duality. With approximately orthogonal attractors, the couplings of the attractor network (left) can be written as a sum over attractors, $J_{ik} = \sum_j \beta_j W_{ij} W_{kj}$. The network is then equivalent to one in which the attractors are explicit "virtual nodes" (right) that pool evidence from many regions and feed the result back, the Hopfield network–restricted Boltzmann machine equivalence (Barra et al., 2012).*

associative memories, patterns remain transiently retrievable above capacity from the slow remnants of their former basins (Clark, 2026). Unlike the spurious states of classical Hopfield networks (Amit et al., 1985), they are useful here: the network can span the whole subspace of learned patterns and generalize to new inputs. Dense associative memories can likewise combine exponential capacity with emergent, useful minima that are not stored patterns (Hoover et al., 2025). Mixing can also unfold in time, as quasi-periodic alternation across several attractors whose time-averaged posterior matches the data. Self-orthogonalization is what makes this work, and it offers a model for oscillatory brain dynamics (Aitchison and Lengyel, 2016; Echeveste et al., 2020).

- **Self-solenoidization** HYPOTHESIS observed in simulations. With our emergent learning rule, sequential presentation yields asymmetric couplings (Sompolinsky and Kanter (1986); Kleinfeld (1986); Figure 2). During transitions, the landscape set by the symmetric couplings tends to absorb the flow components along the free-energy gradient, via a mechanism closely related to self-orthogonalization. What remains is encoded mainly in the antisymmetric couplings, as a transversal flow along the free-energy isocontours: approximately solenoidal flow that replays sequences without changing the steady state (transversality derivation is in progress, full solenoidality is only a conjecture). Self-solenoidization would thus be an on-line, approximate Helmholtz decomposition, as self-orthogonalization is an on-line, approximate principal component analysis. It is the learned counterpart of potential–flux decompositions of non-equilibrium dynamics (Ao, 2004; Kwon et al., 2005; Wang et al., 2008; Yan et al., 2013). Changing only the antisymmetric component may allow new action policies without altering the perceptual model (Section 4.5).
- **Continual learning** HYPOTHESIS initial demonstration PUBLISHED. Two processes support it. First, stochastic multistability causes spontaneous replay of stored patterns, especially but not only in sequence dynamics (Figure 2), which consolidates old memories, as hippocampal and human replay is thought to do (Wilson and McNaughton, 1994; Liu et al., 2019) and as generative replay does in machine learning (Shin et al., 2017; van de Ven et al., 2020). Second, self-orthogonalization stores new information only as its residual relative to what is known, so old knowledge is preserved as long as it remains at least mildly useful. This parallels orthogonal-subspace methods in machine learning (Zeng et al., 2019) and orthogonal task representations in brains and networks (Libby and Buschman, 2021; Flesch et al., 2022), as well as attractor scaffolds in hippocampal models whose memory degrades gracefully rather than collapsing (Chandra et al., 2025). Together they give resistance to catastrophic forgetting (McCloskey and Cohen, 1989; French, 1999) without task labels or explicit regularization (Kirkpatrick et al., 2017). The same two processes should matter on long-horizon tasks, where a system has to keep learning while it acts, without eroding what it learned earlier in the task (Section 1.1; P23).

4.4 Noise as information

In these networks, noise is not a nuisance to be suppressed. Its size encodes the precision of what a unit represents IN PREPARATION partly PUBLISHED.

Each unit's fluctuations sample its posterior, so their variance is the posterior variance and, by the fluctuation–dissipation relation of the exponential family (Kubo, 1966), also the unit's susceptibility, or gain. A unit is therefore most responsive where it is most uncertain, and least responsive where it is saturated and certain: uncertainty is physically available to the system twice, as the size of its fluctuations and as its gain.

In information-geometric terms, this quantity is the Fisher metric (Amari, 1998, 2016), in which prediction-error tangents are measured. Precision weighting is thus simply the geometry of that tangent space: the metric supplies the precision weighting that predictive coding requires (Feldman and Friston, 2010), without a separate mechanism for estimating precision. At the network level, the metric is shaped by learning, through the couplings, and it separates the gradient and rotational parts of the dynamics (Section 4.5). Because finite samples produce exactly this kind of noise, a system that learns from finite experience represents its uncertainty without extra machinery.

This view connects to sampling-based accounts in which neural variability represents uncertainty (Hoyer and Hyvärinen, 2003; Fiser et al., 2010; Buesing et al., 2011; Orbán et al., 2016), to evidence that gain variability tracks uncertainty in visual cortex (Hénaff et al., 2020), and to work on the sources and benefits of noise in neural systems (Faisal et al., 2008; McDonnell and Ward, 2011). It is also what makes the framework a natural fit for stochastic and thermodynamic computing, which treat physical noise as a resource rather than an error (Camsari et al., 2019; Coles et al., 2023; Melanson et al., 2025; Jelinčič et al., 2025).

4.5 Multistability, metastability and intrinsic action

Two dynamical properties of the networks matter both for brain dynamics and for action. Multistability comes from the attractors, and metastability from what the network learns about the transitions between them.

Multistability PUBLISHED. Multiple attractors make the networks multistable, as theory, simulations and our brain models consistently show. Noise is what makes this multistability dynamic: rather than settling in one attractor, the system visits several over time. Such noise-driven exploration of multiple attractors has been proposed as a mechanism of resting-state brain dynamics (Deco et al., 2010; Deco and Jirsa, 2012). A deformable attractor manifold has recently been fitted to resting-state fMRI in the same spirit (Athanasiadis et al., 2025), and the geometry of dynamics on a fitted landscape has been related to attention (Song et al. (2026); see Section 1.3).

Metastability HYPOTHESIS observed in simulations, see Figure 2D. Sequence learning produces asymmetric couplings and a non-equilibrium steady state, with transient dwelling near attractors and directed transitions between them (Kelso, 1995; Rabinovich et al., 2008; Tognoli and Kelso, 2014; Hancock et al., 2024). Because of self-orthogonalization, we expect these dynamics to be dominated by approximately solenoidal flow over the scalar free-energy landscape (self-solenoidization, Section 4.3). Spisak and Friston (2026) give a heuristic explanation of why this should happen; a rigorous derivation is still to be done. Broken detailed balance in human brain dynamics (Lynn et al., 2021; Kringelbach et al., 2024) is consistent with this picture, and the non-equilibrium thermodynamics of networks with asymmetric couplings can be characterized analytically in kinetic Ising models (Aguilera et al., 2023). In cortex, expectation modulates metastable attractor dynamics and speeds up coding (Mazzucato et al., 2019), close to the expectation-as-prior account tested in P6.

Intrinsic action: paths over a learned world model HYPOTHESIS. The intuition is that the network learns paths over a world model it has already learned. Variational free energy (VFE) describes perception on the scalar landscape. A related quantity, possibly the expected free energy (EFE) of active inference (Friston et al., 2015; Parr et al., 2022), would describe intrinsic action, carried by the antisymmetric couplings between pairs of nodes. These couplings induce quasi-solenoidal dynamics: paths along isocontours of the landscape that determine the transitions. Without input, the same dynamics would produce spontaneous fluctuations at rest, or something like dreaming (Hobson and Friston, 2012), as in the spontaneous replay of Spisak and Friston (2026). Whether the standard EFE arguments carry over to these networks is an open question (Millidge et al., 2021). An alternative path-space free energy is the control cost of path-integral control, the relative entropy between controlled and passive path measures (Kappen, 2005; Kappen and Ruiz, 2016).

Decisions and planning HYPOTHESIS. Learned transition paths could support decisions based on learned data, and planning as inference (Attias, 2003; Botvinick and Toussaint, 2012), especially across multiple timescales (Kiebel et al., 2008), in the spirit of predictive maps and replay-based planning (Pfeiffer and Foster, 2013; Stachenfeld et al., 2017). This has a continual-learning side too: the perceptual model, or world model, need not be relearned for every new task. Only the transition paths across its attractors change. The most likely paths between two attractor states are Schrödinger bridges, which can be computed by stochastic optimal control (Chen et al., 2016; Holdijk et al., 2023).

Asymmetry, rotation and directed flow HYPOTHESIS. Linearizing the mean-field dynamics around a fixed point separates, in the Fisher metric of Section 4.4, a gradient-like part carried by the symmetric couplings from a rotational part carried by the antisymmetric ones. When asymmetry dominates, the local dynamics contain damped rotations, which links asymmetry directly to oscillation. Asymmetry is thus the minimal coupling structure that distinguishes temporal consequence from static correlation, and it turns an attractor landscape into directed sequence dynamics. Thermodynamically, the gradient and rotational parts correspond to excess and housekeeping entropy production (Hatano and Sasa, 2001). In the dual picture (Section 4.3), attractors act as hidden nodes, and learned sequences live in directed couplings between them. If region-to-region asymmetries mainly serve these transitions, the antisymmetric couplings are low-rank and confined to the attractor subspace: the N(N - 1)/2 free asymmetries between region pairs reduce to about m(m - 1)/2 flows between m attractors. Such flows can leave the stationary density of the landscape unchanged, that is, be solenoidal, even though the attractors differ in depth, provided that each transition is weighted by the precision of the attractor it targets, or that the transitions form closed cycles through three or more attractors. Both routes lead to the same testable signature: attractor occupancies set by the symmetric landscape alone, with asymmetric currents that leave them unchanged. Finding this would itself be evidence of approximate self-solenoidization (P9). The derivations are sketched in *Technical sketch: asymmetry, rotation and solenoidal flow.*

Technical sketch: asymmetry, rotation and solenoidal flow
Gradient and rotation. The mean-field dynamics and their linearization around a fixed point make the split between gradient and rotation explicit:

$$\dot{\theta}_i = -\gamma\Big[\theta_i - b_i - \sum_j W_{ij}\, L_j(\theta_j)\Big], \qquad \delta\dot{\boldsymbol{\theta}} = M\,\delta\boldsymbol{\theta}, \qquad M = -\gamma\,(I - WD), \qquad D = \operatorname{diag}\left(\psi_1''(\theta_1), \ldots, \psi_n''(\theta_n)\right) \tag{9}$$

Here D holds each unit's susceptibility, which is also its variance and its Fisher information (Section 4.4). Splitting the couplings into symmetric and antisymmetric parts, W = S + A, and lowering an index with the Fisher metric separates a symmetric, gradient-like term from an antisymmetric, rotational one:

$$DM = -\gamma\,(D - DSD) + \gamma\,DAD \tag{10}$$

For two units with J12 = S + A and J21 = S - A, the eigenvalues are $-\gamma \pm \gamma\sqrt{((S^2 - A^2)\,\psi 1''\,\psi 2'')}$. When |A| >|S|, the local dynamics contain a damped rotation with frequency $\omega = \gamma\sqrt{((A^2 - S^2)\,\psi 1''\,\psi 2'')}$, which links asymmetry directly to oscillation. Symmetric couplings can admit a reversible, equilibrium model, whereas antisymmetric couplings generically obstruct that interpretation and support directed probability currents.

Solenoidal flow through the attractors. With the m attractor patterns as the columns of Ξ, the N(N - 1)/2 free asymmetries between region pairs reduce to a small matrix Ω of attractor-to-attractor flows, with about m(m - 1)/2 free entries. With Λ holding the precision (curvature) of each basin, a linear–Gaussian sketch with isotropic noise gives the condition under which this flow is solenoidal, that is, leaves the stationary density of the landscape unchanged (Kwon et al., 2005):

$$H = \Xi\,\Lambda\,\Xi^\top, \qquad T = \Xi\,\Omega\,\Xi^\top, \qquad \text{solenoidal} \iff \Omega = Q\,\Lambda, \quad Q^\top = -Q \tag{11}$$

Exact solenoidality therefore does not require attractors of equal depth, which the present models do not have. It requires each directed coupling to be weighted by the precision of the attractor it targets, $\Omega\mu\nu = Q\mu\nu\ \lambda\nu$: an evidence weighting of the transitions. Besides their antisymmetric part, the transition couplings then carry a small symmetric part, $\frac{1}{2}\ Q\mu\nu\ (\lambda\nu - \lambda\mu)$, which grows with the difference in depth. Whether the predictive-coding learning rule produces this weighting by itself, since it subtracts precision-weighted predictions, is open, and simulations can measure it directly. A second route appears at the level of metastable hopping between attractors. There, added probability currents leave the occupancies unchanged exactly when they decompose into closed cycles, each added transition rate being scaled by the inverse occupancy of its source (Schnakenberg, 1976). A directed flow between just two attractors cannot preserve occupancies, but a cycle through three or more can, whatever their depths.

An opportunity for directed connectivity from fMRI (HYPOTHESIS). Estimating directed connectivity between single pairs of regions from fMRI is notoriously hard, whether with dynamic causal modelling, Granger causality or related methods (Friston et al., 2003; Smith et al., 2011; Seth et al., 2015; Gilson et al., 2016), and the antisymmetric component is typically weak compared with the symmetric one. If asymmetries serve attractor transitions, estimating them at the attractor level pools evidence from many region pairs into a few attractor-to-attractor flows, which should improve statistical power; where needed, the flows can be projected back onto region pairs (P22). This applies the logic of the sufficiency test (Section 5.2) to directed structure, and it fits the coarse-grained state spaces in which broken detailed balance has been detected in human brain dynamics (Lynn et al., 2021). Region-specific haemodynamic delays can create spurious asymmetries that do not average out at the attractor level; deconvolution, simulations with region-specific haemodynamic responses, and sequence tasks with a known order serve as controls.

5. Attractors as levels of description

From the observer to the system. In Section 4.2, coarse-graining was done by an observer, who chose the slow observables. Collective dynamics suggest another source: attractors are, in effect, how a system coarse-grains itself, creating new, slow, macroscopic variables from within.

5.1 Attractors are not epiphenomena

A common objection to attractor accounts of brain function is that attractors are epiphenomena: convenient summaries of the dynamics, while the causal work is done by neurons and synapses alone. In this programme they are instead how new timescales, and with them new levels of description, come into being HYPOTHESIS builds on PUBLISHED results. Pressed to its end, the objection becomes a philosophical question about causation across levels of description (Kim, 1998), which we do not claim to settle. The programme does, however, offer several concrete reasons for the opposite view, each of which can be tested:

- **Attractors create timescales.** An attractor is where the dynamics slow down: the system typically dwells near it far longer than any of its units takes to relax. It is therefore a new, slower variable that the system creates for itself, and this is exactly the kind of timescale separation from which Section 3 derives perception and learning. The state of an attractor network can then act as a slowly varying field, or prior, for faster processes, so that a construction of the same kind can repeat one level up (A2, Section 8.1). Attractors are thus not a by-product of inference and learning. They are how a system produces the separation of timescales on which its inference and learning rest. Hierarchies of intrinsic timescales across the cortex are consistent with this view (Kiebel et al., 2008; Murray et al., 2014), as is the self-organization of functional hierarchies in recurrent networks with multiple timescales (Yamashita and Tani, 2008).
- **Attractors do computational work.** They can store priors and separate latent causes through self-orthogonalization; they may bridge delays in trace conditioning and working memory; and, through their transitions, they can carry sequences and intrinsic action (Sections 4.3–4.5 and 6). With symmetric couplings and approximately orthogonal attractors, the network is statistically equivalent to one in which the attractors are explicit hidden nodes that pool evidence and feed it back (Barra et al., 2012) (see Section 4.3). A description in which attractors are variables is then not a metaphor but an equivalent model.
- **Attractors make a difference under intervention.** On the interventionist account of causation (Woodward, 2004), a variable is a cause if intervening on it changes what follows. Perturbation experiments reveal attractor dynamics in exactly this sense. An activity bump imposed optogenetically on the fly's heading circuit is maintained by the circuit (Kim et al., 2017). Persistent activity in mouse frontal cortex returns to discrete attractor states after transient perturbations, or occasionally switches to the other attractor, followed by a switch in the animal's action (Inagaki et al., 2019), and attractor dynamics gate the influence of distracting input on decisions (Finkelstein et al., 2021). The programme's own predictions are stated at the attractor level (extinction adds an attractor, chronic pain deepens one, expectations shift the posterior between them; P5, P6, P8) and can be probed with current neuroscience techniques, e.g. in closed loop with real-time fMRI (B6).
- **Attractors can form closed levels.** A macroscopic level is self-contained if it predicts its own future as well as the full microscopic state does (informational closure), and if interventions on macroscopic variables suffice to determine macroscopic outcomes (causal closure). Rosas et al. (2024) formalize such levels, which they liken to software running on physical hardware, and show that in a Hopfield network the overlaps with the stored patterns form a closed level of this kind: memory retrieval is computed at the level of the attractors, independently of microscopic detail (Hoel et al., 2013; Rosas et al., 2020). Timescale separation is what makes slow variables approximately closed: in synergetics, slow order parameters are generated by the fast variables and in turn enslave them, a circular causality in which the macro level is anything but idle (Haken, 1983; Kelso, 1995). Section 4.2 adds that the inferential form survives coarse-graining, so a closed level can itself infer and learn. Whether attractor-level descriptions of brain dynamics are closed in this sense is an empirical question (P17). If it is, closure has a practical corollary: for whatever is computed at that level, the attractor state is a sufficient statistic, which turns closure into a test of what is computed in the attractors (Section 5.2; P21).

Simon (1962) argued that complex systems evolve far more readily when they can build on stable intermediate forms, and that the result is a nearly decomposable hierarchy, in which fast interactions within subsystems

are separated from slow, aggregate interactions between them. Attractors are dynamical counterparts of such stable intermediate forms. Where levels are approximately closed in this sense, each can be studied largely in its own terms, without simulating everything beneath it, and an elementary step that recurs at each level needs to be understood only once (Section 11).

5.2 What is computed in the attractors? Counterfactuals and a sufficiency test

For brain data, the programme aims at mechanism, not at predictive accuracy. Two kinds of test follow: counterfactual simulation, and a test, derived from closure, of which quantities are carried by large-scale attractors and which are computed below them (HYPOTHESIS) proposed method.

Mechanism, not accuracy. Data-driven predictive models of brain and behaviour are optimized to predict, and a mechanistic model need not beat them at it. Its value lies in answering questions they cannot answer: what would happen under an intervention that is absent from the training data. The fitted networks are generative and can be manipulated in silico: an attractor can be deepened or flattened, prior precision raised or lowered, the biases of selected regions shifted, or couplings altered. The decisive test is whether such counterfactual simulations, specified in advance, predict the effects of real manipulations, such as placebo and nocebo instructions, conditioning and extinction, pharmacological challenge, non-invasive brain stimulation or closed-loop real-time fMRI. The comparison is with mechanistically different variants of the same model (couplings fixed, biases only, a landscape without attractor structure), not with black-box predictors. Conventional predictive models keep an important role, however: as shown next, they set the reference that an attractor-level description has to reach, not beat.

Two levels in one fit. Score matching separates two components of the regional dynamics (graphical abstract, Line B). The couplings define the landscape and its attractors: dynamics at the network level. The biases capture what drives each region from below and from outside: mesoscale computation within regions, and sensory input. Both can be read out as time series from the same data: the attractor state a(t), given by the overlaps of the regional state x(t) with the fitted attractors (or by the attractor towards which each state relaxes), and the input b(t) inferred by the fitted model.

Closure implies sufficiency. If the attractor level is closed in the sense of Section 5.1 (P17), and a quantity y (a perceptual report, a behavioural choice, an experimental condition or a clinical variable) is computed at that level, then the regional data carry no information about y beyond what the attractor state carries. The attractor state is then a sufficient statistic for y:

$$p\big(y \mid x\big) = p\big(y \mid a(x)\big) \quad \Longleftrightarrow \quad I(y;x) = I(y;a) \tag{12}$$

Operationally, y should be predictable from the attractor time series just as well as from the full regional data: not better, but without loss. Because a(t) and b(t) are both computed from the regional time series by the fitted model, the chain rule of mutual information splits what the data carry about y exactly into three parts:

$$I(y;x) = \underbrace{I(y;a)}_{\text{attractor level}} + \underbrace{I(y;b \mid a)}_{\text{upstream input}} + \underbrace{I(y;x \mid a,b)}_{\text{neither}} \tag{13}$$

Quantities for which the first term exhausts the total are computed in attractor space. Those that need the second term are carried by upstream, mesoscale processing or by sensory drive. A substantial third term points to structure that the model misses. In this way closure, a property of a level, becomes a criterion for what lives on that level.

Mixed cases and controls. Many brain functions will be mixed, and the attractor share I(y; a)/I(y; x) is then a graded measure rather than a verdict. Its value is that it is principled rather than descriptive: the criterion follows from the theory, not from choosing whichever feature set predicts best. Three controls keep it honest. First, sufficiency must be specific: equally compact coarse-grainings that ignore the attractors (principal components, random projections, canonical networks of the same dimension) should lose information that the attractor coordinates keep; otherwise the result reflects compression in general rather than attractors. Second, the information terms are estimated from the cross-validated performance of nested models fitted with the same learner, and are bounded in practice by sample size, model capacity and haemodynamic blurring. Equal performance is therefore established with equivalence tests, not read off a non-significant difference. Third, sufficiency for an external quantity is necessary but not sufficient for closure, which also

requires the attractor level to predict its own future (P17), so the two tests are run together. The same logic applies between resolutions: replacing x with data from a finer atlas, or at voxel level, asks whether the coarse attractors suffice for what the finer data carry (Section 4.2).

An example: pain. Pain is a natural test case because it combines strong input with strong expectations. If expectations act as attractor priors (P6), placebo and nocebo effects on reported pain should be largely attractor-sufficient, and an in-silico shift of the corresponding prior should reproduce them. The effect of stimulus intensity, by contrast, should be carried mostly by the biases, as nociceptive drive. In chronic pain the same decomposition can be followed over time, to see whether chronification moves pain from an input-carried to an attractor-level phenomenon, or the reverse (P8).

6. Emergent, biology-like phenomena: hypotheses

Several well-known features of biological learning may follow from the same few properties, above all from self-orthogonalization. Below we highlight a few examples, each a hypothesis to be tested rather than a result. Further-reaching possibilities are collected in Section 11.

- **Primacy effect** HYPOTHESIS. Because new information is stored only as a residual relative to existing attractors, early material serves as the reference frame for what follows and should be remembered better. This would mirror the primacy effect in human memory (Murdock, 1962), in contrast to classical palimpsest networks, which forget their oldest memories first (Nadal et al., 1986; Parisi, 1986).
- **Extinction, renewal and reinstatement** HYPOTHESIS. Self-orthogonalization restricts how representations can change:
  - *Acquisition.* A newly learned representation is stored in an attractor approximately orthogonal to all previous ones, so only its residual is stored, and old representations shift as well (remapping) (Colgin et al., 2008).
  - *Context.* Overlap with old memories, for example an association with fear or pain, induces correlations that must be balanced by non-overlapping components to keep the attractors orthogonal. This is how context A becomes part of the memory.
  - *Extinction.* The old memory could only be erased if everything else stayed unchanged, which is practically impossible. Instead a new attractor forms, again orthogonal to the others. Because the stimulus is similar or identical but the outcome differs, self-orthogonalization puts strong pressure on using context to tell the two situations apart: context B.
  - *Renewal and reinstatement.* The old context, or the unconditioned stimulus itself, therefore brings back the original response. Through sequence learning, the network may even anticipate it.
  - *Related work.* This mechanistic account parallels the evidence that extinction is new, context-dependent learning (Rescorla and Heth, 1975; Bouton, 2004; Rescorla, 2004), latent-cause models in which extinction creates a new latent cause (Gershman et al., 2010, 2017), state-splitting accounts of renewal (Redish et al., 2007) and the neurobiology of fear extinction (Milad and Quirk, 2012).
- **Trace conditioning and working memory** HYPOTHESIS. Self-sustaining attractor activity that persists after the stimulus has gone would bridge the interval in trace conditioning (Clark and Squire, 1998) and support working memory, as in classical persistent-activity models (Goldman-Rakic, 1995; Amit, 1997; Wang, 2001). Activity-silent accounts (Stokes, 2015) provide the contrast to test against.
- **Oscillations during acquisition** HYPOTHESIS. When a new stimulus is not well explained by any existing attractor, ghost attractor or superposition, only oscillatory sampling across attractors remains. This is slow and costly, because the boundaries between attractors have high free energy, so learning will try to store the new information in an attractor, or at least a ghost, while keeping orthogonality (remapping). The oscillations should then fade as learning proceeds. This may relate to theta-band activity during acquisition and encoding (Buzsáki, 2002; Hasselmo et al., 2002; Seidenbecher et al., 2003; Likhtik et al., 2013), and to the oscillations that emerge in circuits optimized for sampling-based inference (Aitchison and Lengyel, 2016; Echeveste et al., 2020).
- **Curiosity and exploration** HYPOTHESIS open. How far the usual active-inference arguments for epistemic value (Friston et al., 2015, 2017) carry over to these networks is unclear, but uncertainty here naturally encourages exploration (Oudeyer, 2007; Schmidhuber, 2010; Gottlieb and Oudeyer,

2018). In this picture a "dark room" (Friston and Ao, 2012) is not predictable: it is a lack of evidence, which leaves spontaneous activity to sample the priors and may produce hallucination-like states. This links to accounts of hallucinations as strong priors (Powers et al., 2017; Corlett et al., 2019) and to pain without input, as in phantom-limb pain (Flor et al., 2006).

- **Reinforcement learning** HYPOTHESIS. The networks are compatible with reinforcement learning in the way active-inference accounts propose (Friston et al., 2009; Sutton and Barto, 2018), but this relies on fixed priors set by evolution, such as pain. Pain research is therefore a good test bed (Büchel et al., 2014; Wager and Atlas, 2015; Tabor et al., 2017).

7. Predictions and falsification criteria

Every prediction below is stated in advance, with the test bed and the result that would count against it. Self-orthogonalization is the system's most prominent signature, but it is hard to test in high dimensions: random vectors also tend towards orthogonality. Real inputs are almost always structured and correlated, so orthogonality of learned attractors is not trivial. Still, the decisive test we propose is the projector-network signature (P2), not orthogonality alone. All geometric tests use strong null models: phase-randomized time series, symmetry-preserving permutations, covariance-matched Gaussian models, and condition-label shuffling. Analysis plans will be preregistered (see the note at the start of Section 8). For the neuroscience predictions the criterion is mechanistic: counterfactual simulations specified in advance, and the sufficiency test of Section 5.2, rather than predictive accuracy beyond that of existing data-driven models.

#	Prediction	Test bed	Would count against it
P1	Learned attractors are approximately orthogonal, even for correlated inputs	Rest and task fMRI; simulations	Attractor angles indistinguishable from matched null models
P2	Decisive test of self-orthogonalization: the networks approach a Kanter–Sompolinsky projector network, whose attractors coincide with the leading eigenvectors of the coupling matrix	fMRI-fitted networks; simulations	Eigenvector–attractor alignment no better than null models
P3	Orthogonalization depends on the timing of the learning rule: pairing the error with the renewed state orthogonalizes; pairing it with the preceding state merges attractors	Simulations; physical hardware	Orthogonalization with reversed pairing, or none with the correct pairing
P4	Noise encodes precision: a unit's variance equals its gain, and variability tracks inferred precision	Physical hardware; neural and BOLD variability vs responsiveness	No relation between variability and gain or precision
P5	Extinction creates a new, approximately orthogonal attractor rather than erasing the old one; renewal is a context-driven return to it	Threat-learning and extinction fMRI; benchmark battery	Extinction as weakening of the original attractor; renewal not predicted by the fitted landscape
P6	Expectations act as attractor priors: placebo and nocebo shift the posterior towards expectation-congruent states, scaling with prior precision	Placebo and nocebo fMRI	An in-silico shift of the prior does not reproduce the observed placebo and nocebo effects, or these effects are not attractor-sufficient (P21)
P7	Task activity decomposes into network-level (couplings) and input-driven (biases) components	Task fMRI; 7T laminar fMRI (feedback vs feedforward layers)	Decomposition does not generalize across tasks or maps onto the wrong layers
P8	Chronic pain reflects a deepened, harder-to-leave pain attractor (macro-scale) and/or region-intrinsic change (meso-scale), distinguishable within subjects over time by the attractor–input decomposition (P21)	Longitudinal phantom-limb pain cohort	Neither component tracks chronification

#	Prediction	Test bed	Would count against it
P9	Self-solenoidization: sequential experience produces asymmetric couplings whose flow is approximately solenoidal (along free-energy isocontours) and replays sequences in their learned direction, while attractor occupancies stay those set by the symmetric landscape (Section 4.5)	Sequence-learning fMRI (Helmholtz decomposition of fitted flows); simulations	Transition flow mainly along the free-energy gradient, no directed replay, or occupancies shifted by the asymmetric couplings
P10	Prior mixing: inputs outside the stored set are represented by superpositions of attractors, as mixed or ghost attractors or as quasi-periodic alternation whose time average matches the posterior	Simulations; task fMRI; oscillatory dynamics in electrophysiology via collaborators	Time-averaged dynamics no closer to the input than the nearest single attractor
P11	The networks learn continually with limited forgetting	Benchmark battery (Line C)	Forgetting comparable to or worse than naive sequential training
P12	Learning emerges natively from physical dynamics plus the local rule	Thermodynamic, memristive or other emerging hardware	No learning, or learning without the predicted representations
P13	Learning new transitions over familiar states changes mainly the antisymmetric couplings, leaving the symmetric part and its attractors largely intact	Simulations; fMRI-fitted couplings before and after sequence learning	Symmetric couplings change as much as antisymmetric ones
P14	Primacy: early-learned, correlated patterns are retained better than later ones	Benchmark battery; behavioural experiments with correlated material	Recency, or no serial-position effect, for correlated patterns
P15	Acquiring stimuli that existing attractors explain poorly triggers transient oscillatory alternation between attractors, which fades as a new attractor forms	Simulations; EEG or intracranial recordings during conditioning, via collaborators	No oscillatory signature, or one unrelated to how well stimuli are explained
P16	Without input, spontaneous dynamics sample the learned priors: visit statistics at rest reflect the attractors and their learned probabilities (Berkes et al., 2011)	Rest and task fMRI; simulations	Spontaneous statistics unrelated to learned or evoked structure
P17	Attractors form an approximately closed level: the attractor-level state predicts its own future nearly as well as the full regional state, and better than equally compact coarse-grainings that ignore the attractors (Rosas et al., 2024)	Rest and task fMRI; simulations	Attractor-level descriptions no more closed than matched coarse-grainings of equal dimension
P18	The same landscape at lower resolution: attractors fitted at a fine functional atlas and projected onto a coarser, nested atlas match the attractors fitted there directly, and higher resolution splits coarse attractors into descendants that average back to them	Nested multi-resolution atlases; population and precision fMRI	Low match after projection, or unrelated attractor sets at different resolutions
P19	Spatially scattered, size-matched random parcellations lose the structured attractors and collapse towards a single global mode; contiguous random parcellations degrade gradually, the finest attractors first	The same data, randomly parcellated	Random parcellations preserve the attractors as well as functional ones do
P20	Orthogonality is inherited: in the size-weighted parcel metric, the inner products of coarse attractors match those of the projected fine attractors, better than dimension-matched nulls do	Nested atlases; simulations with known ground truth	Coarse inner products unrelated to the projected fine ones, or no better than nulls

#	Prediction	Test bed	Would count against it
P21	Attractor sufficiency (corollary of P17): quantities computed at the attractor level are predicted from the attractor time series as well as from the full regional data, and better than from equally compact coarse-grainings that ignore the attractors; what the attractors miss is carried by the fitted biases (Section 5.2)	Task and rest fMRI with perceptual, behavioural and clinical targets (e.g. pain, placebo); simulations in which the level of computation is known	Attractor coordinates no more sufficient than matched coarse-grainings for any target; the test fails to recover the known level in simulations; or what the attractors miss is not carried by the biases either
P22	Directed flow through attractors: asymmetries in fitted couplings are concentrated in the attractor subspace, so attractor-to-attractor flows detect directed structure with more power than pairwise estimates, and sequence tasks reveal flows in the learned direction (Section 4.5)	Rest and sequence-task fMRI; simulations with known directed couplings and region-specific haemodynamics	Asymmetries no more concentrated in the attractor subspace than in random subspaces of equal dimension; no power gain over pairwise estimates in simulations with ground truth; or flows explained by haemodynamic delays
P23	Long-horizon advantage: on long, open-ended tasks, networks of this class keep improving beyond the gains of repeated sampling from their own frozen state, and overtake conventional models (deep networks, language-model agents) that perform equally well at the start HYPOTHESIS	Benchmark battery (Line C): long-horizon tasks with a repeated-sampling baseline and initially matched controls	Gains no better than repeated sampling, or no advantage over initially matched conventional models

Negative results are informative by design: they delimit where the construction applies and point to mechanisms missing from the theory, and they will be published.

8. Technical plan

A plan, to be extended by preregistrations
This section is an overview. It will be extended by detailed appendices that serve as preregistrations of the methodology. Each will be released with its own DOI before the corresponding work begins, so that methods and analysis plans are fixed in advance and readers can check what was done against what was planned.

The three lines share one model class, one set of definitions and one set of predictions. Results from each line are checked against the others, and are often published together. Sharing one model also creates synergies that neither line could produce on its own (Figure 6). The same Bayesian reading of the attractors, in which priors and inputs are combined according to their precision, applies to task fMRI in Line B and to artificial data, such as noisy faces, in Line C. A finding in one domain therefore translates directly into a hypothesis or a benchmark in the other: brain data show which regimes of prior and input precision biology uses, and machines test those regimes under full control.

8.1 Line A – Approach and theory

- **A1. Complete and test the unit-level construction.** Publish the derivation; test its contingent predictions in finite jump-process simulations in which no inference or learning rule is coded: the form of the nonlinearity under different resting laws, fluctuation–gain equality, and the dependence of orthogonalization on learning-rule timing (P3, P4).
- **A2. Composition and hierarchy.** Formalize how evidence at one level acts as the effective field at the next, and how nested timescales give rise to hierarchical message passing. Derive principled coarse-graining criteria: bounds on how well attractors are preserved as a function of within-parcel heterogeneity, and conditions under which the coarse transition law, not only the attractor set, is preserved (Section 4.2). Show how attractors at one level supply the slow timescale for the next, and characterize when the resulting levels are informationally and causally closed (Section 5.1; P17).

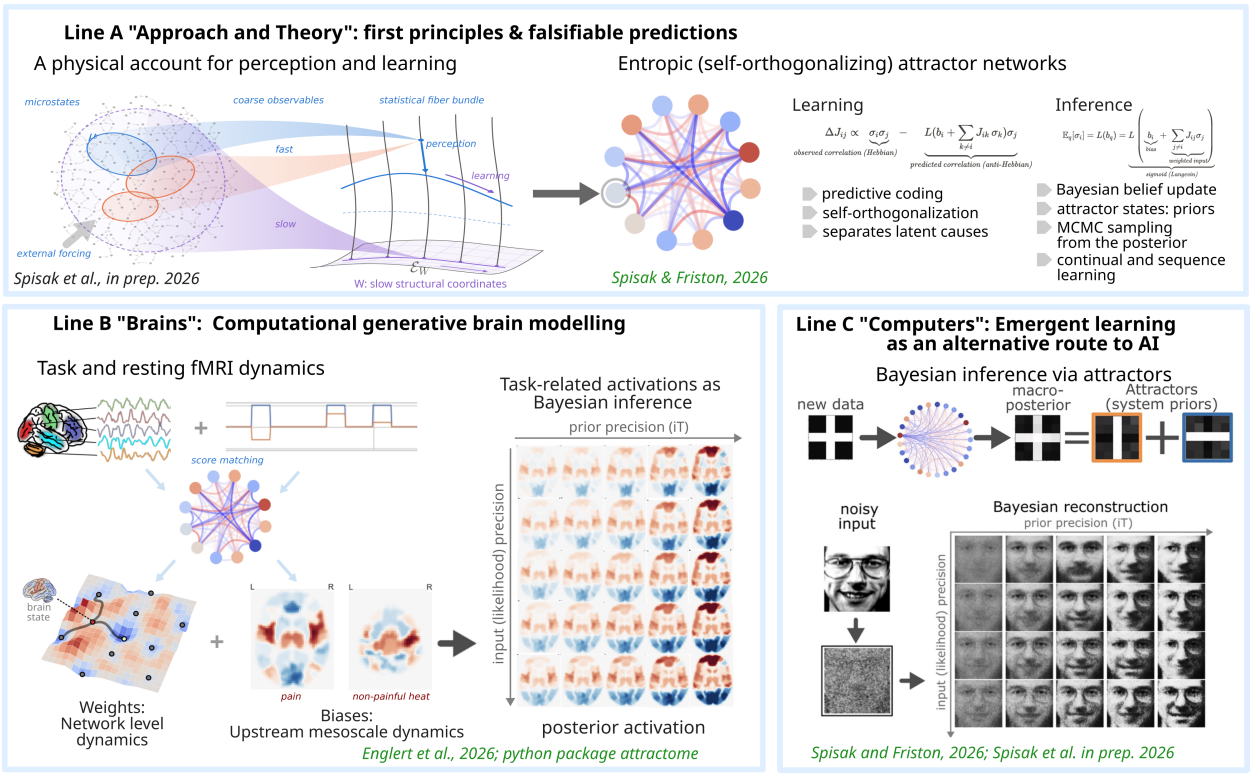


Figure 6: ***One model for brains and machines (the graphical abstract, repeated). Line A*** *The theory: a minimal unit in which fast inference and slow learning arise from coarse observables of hidden microstates (Spisak et al., in preparation), composed into entropic, self-orthogonalizing attractor networks with local inference and learning rules (Spisak & Friston, 2026). The same networks serve both other lines.* ***Line B*** *Fitted to task and resting-state fMRI by score matching, the networks separate network-level dynamics (couplings and their attractor landscape) from upstream, region-intrinsic dynamics (biases; here for painful and non-painful heat). Read as Bayesian inference, task-related activations change systematically with prior and input precision (Englert et al., 2026; attractome).* ***Line C*** *Trained on artificial data, the same networks perform Bayesian inference via attractors: the posterior combines new data with the attractors as system priors, and noisy face images are reconstructed with a trade-off between prior and input precision that mirrors the brain data in Line B (Spisak & Friston, 2026; Spisak et al., in preparation). Because the three lines share one model, a result in one translates directly into a hypothesis or benchmark for the others.*

- **A3. Action and planning.** Develop the network-level account of action: solenoidal flow from antisymmetric couplings as intrinsic action, and effector-driving nodes as extrinsic action. Study planning as sequence prediction over the learned landscape, and the link between learnable surprise and exploration.
- **A4. Thermodynamic cost.** Relate the quantities of the construction (alignment gap, learnable versus unlearnable surprise) to the heat dissipated during perception and learning, as a bridge to physical implementation. Because each synapse is the thermodynamic Legendre conjugate of a correlation, learning has a thermodynamic price that can be computed in theory and measured in hardware. Learning is a slow protocol W(t) whose changes of ψ are free-energy differences, so the nonequilibrium work relations apply (Jarzynski, 1997; Crooks, 1999); under slow driving, the dissipated work is a thermodynamic length in a friction metric related to the Fisher metric (Sivak and Crooks, 2012), which ties the cost of learning to the separation of timescales.

Methods: non-parametric information geometry, stochastic thermodynamics (Seifert, 2012; Welling et al., 2026), averaging and adiabatic theory, and large-scale simulation.

8.2 Line B – Brains: generative models of large-scale brain dynamics

- **B1. Model fitting.** With the open-source package *attractome*, fit networks to resting-state, task and naturalistic fMRI by score matching with haemodynamic deconvolution, estimating couplings

(network-level dynamics) and biases (upstream, mesoscale input). Validate generatively, by simulating held-out data, and against null models. This includes individual-level models from dense, repeatedly sampled single-subject datasets. Before interpretation, the fitted landscapes are tested across resolutions: project and match across nested atlases, measure homogeneity, and compare with random parcellations, in population and single-subject data (Section 4.2; P18–P20).

- **B2. Tests of the theory.** Attractor orthogonality and eigenmode alignment (P1, P2), the couplings–input decomposition (P7), self-solenoidization, transition learning and directed attractor-to-attractor flows in sequence tasks (P9, P13, P22), prior mixing and oscillatory alternation between attractors (P10, P15), spontaneous sampling of priors at rest (P16), and the closure of the attractor-level description (P17) and its corollary, attractor sufficiency (P21). Basin depths and occupancies in the fitted landscapes are estimated with multistate free-energy methods (Holdijk et al., 2026a). Ultra-high-field (7T) laminar fMRI tests whether inferred priors and inputs map onto feedback and feedforward cortical layers.
- **B3. Learning without forgetting: extinction and renewal (P5).** Fit landscapes before and after acquisition, extinction and renewal in threat-learning paradigms, and test whether extinction adds a new, approximately orthogonal attractor.
- **B4. Expectations as priors (P6).** Model placebo and nocebo effects on pain as precision-weighted combinations of attractor priors and nociceptive input, using placebo-imaging consortium data. Test whether in-silico shifts of the prior, specified in advance, reproduce the observed effects, and whether placebo responses are attractor-sufficient while the effects of stimulus intensity are carried by the input (Section 5.2).
- **B5. Plasticity in pain chronification (P8).** Separate meso-scale (region-intrinsic) from macro-scale (network) plasticity in chronic low back pain, fibromyalgia and longitudinal phantom-limb pain cohorts, using the attractor–input decomposition to follow, within subjects, the level at which pain is carried as it becomes chronic (P8, P21).
- **B6. From predictive to mechanistic neuromarkers.** Brain-based predictive models of pain, cognition and clinical outcomes have matured: they can be replicable with moderate sample sizes (Spisak et al., 2023), and some generalize to independent patient populations (Spisak et al., 2020). Most, however, rest on data-agnostic machine learning, so their behaviour out of context is hard to predict, their features are hard to map onto mechanisms, and they cannot simulate what has not been measured. The aim here is not to outperform them but to explain what they capture. The sufficiency test (Section 5.2) asks whether the information a predictive model exploits is carried by large-scale attractors or by upstream input, with the predictive model itself as the reference to be matched. Compact model-based features (attractor depth, dwell times, transition costs (Holdijk et al., 2023)) are validated externally across cohorts, and counterfactual simulation generates intervention hypotheses that are specified in advance. Real-time fMRI with reinforcement learning from brain feedback provides closed-loop, causal tests of these hypotheses.

Data: large open datasets and cohorts from established collaborations (placebo and pain consortia, two DFG collaborative research centres, an ERA-NET pain consortium).

8.3 Line C – Computers: emergent learning as an alternative route to AI

- **C1. Software and scaling.** Efficient, open implementations; map the networks' operational regimes (temperature, noise, input precision) and scaling behaviour.
- **C2. A neuroscience-inspired benchmark battery.** Continual learning, extinction, reinstatement and renewal, interference, primacy, reversal and sequence tasks, matched to the paradigms of Line B so that brain and machine are compared on the same tasks (P5, P11, P14), and long-horizon tasks scored against a repeated-sampling baseline (P23). Baselines: standard continual-learning methods (regularization- and replay-based) and backpropagation-trained recurrent networks.
- **C3. Uncertainty and generative inference.** Bayesian reconstruction and denoising as a function of prior and input precision; calibration of the networks' uncertainty; generalization through prior mixing (P10).
- **C4. Native physical implementation (P12).** Stochastic binary units are the tanh case of the minimal unit, which makes probabilistic-bit and related hardware a natural fit. Steps: simulate hardware-constrained networks; run inference natively on hardware with the local learning

rule applied from the measured states; then co-design slowly adapting couplings native to the substrate. Candidate substrates: thermodynamic, probabilistic-bit, memristive, nanomagnetic and related emerging architectures (Camsari et al., 2019; Coles et al., 2023; Harabi et al., 2022; Jelinčič et al., 2025; Holdijk et al., 2026b), with partners. Device evaluation has so far relied on emulation metrics that rate how faithfully a device reproduces biological synapses, neurons or networks (Bruno et al., 2026). As a complement, the construction suggests theory-derived figures of merit: the separation of timescales between state and conductance, a bounded, maximum-entropy response, noise variance that tracks gain, bidirectional local updates with the prescribed timing, support for asymmetric couplings, and, at the network level, attractor orthogonality and resistance to forgetting.

- **C5. Closed-loop agency.** Maze, reversal and detour tasks probe world-model formation, planning and autonomous action, and bridge back to behavioural neuroscience.

9. Risks, limitations and fallbacks

The programme is designed so that each risk, if it materializes, still produces an informative result.

Risk	Consequence	Fallback or what is learned
Assumptions of the construction do not hold for a given substrate	The inferential reading is not guaranteed there	Delimits the scope of the theory; the network models remain usable as phenomenological generative models
Orthogonality in brain data is explained by high-dimensional chance or measurement artefacts	P1 and P2 fail	Strong null models detect this; attention shifts to dynamical predictions (P5–P9)
fMRI is too slow or too noisy for some signatures	Weak tests in Line B	Laminar 7T fMRI, dense single-subject sampling, and simulations with matched noise; electrophysiology via collaborators
Counterfactual simulations fail to predict the effects of real manipulations	The fitted landscapes describe brain dynamics without explaining them	Model variants with fixed couplings or biases locate which component fails; the sufficiency analysis still maps which quantities are carried at the attractor level; results are published either way
Networks do not scale on conventional hardware	Line C limited to small and medium problems	Focus on properties (continual learning, calibrated uncertainty) rather than scale; emerging hardware as the long-run route
Emerging hardware cannot realize slow adaptive couplings	Native learning (C4) not reached	Hardware-in-the-loop learning remains publishable; results inform hardware development
Brain and machine share no common signatures	The common-principle hypothesis fails	A clear negative result: the theory may describe a viable route to learning that brains do not use
Attractor-level descriptions of brain data are not closed (P17 fails)	Attractors may be better treated as descriptions than as a level in their own right	The fitted networks remain generative models of regional dynamics; closure is then tested at finer scales and in simulations, where timescale separation is larger (A2)
Few quantities are attractor-sufficient; most are mixed (P21)	No clean assignment of functions to levels	The attractor share remains a graded, theory-derived measure; a consistently large input share would itself show that large-scale attractors carry less of the computation than proposed
Attractors do not match across resolutions (P18 or P19 fails)	fMRI-fitted networks cannot be read as views of one landscape	Claims are restricted to resolutions validated by held-out generative tests; precision fMRI, 7T data and simulations with known ground truth locate where homogeneity fails

Limitations and open questions

Much of this programme rests on heuristics that still have to be made rigorous. We list the main ones so that readers can weigh them.

- **Sufficient, not necessary.** The unit-level construction shows one route from physics to inference and learning. It does not show that brains, or any particular material, take this route.
- **Strong assumptions.** Timescale separation, bounded coordinates and leading-order interactions are idealizations. Where they fail, the inferential reading is not guaranteed.
- **Heuristics not yet derived.** Self-solenoidization, the duality with antisymmetric couplings, prior mixing through oscillatory alternation, and the role of expected free energy in intrinsic action are hypotheses, not results.
- **Not an exact gradient.** The network learning rule is not the exact gradient of a single objective, so its stationary behaviour is characterized only approximately.
- **Minimal units.** A one-dimensional unit is a coarse-graining of a richer system, not a claim that neurons or regions are one-dimensional. Faster intrinsic dynamics shape its response function, slower ones become its prior and structure; only dynamics on the unit's own timescale are lost, as a measurable memory term. Where this loss matters, multidimensional or spiking units extend the construction.
- **Measurement.** fMRI is slow, indirect and confounded by vascular and physiological signals. Results depend on coarse-graining choices, since a parcellation shows only distinctions that live across parcels (Section 4.2), and orthogonality in high dimensions needs strong null models.
- **Scale.** Demonstrations so far are small to medium in scale. We make no claim of competitiveness with large deep-learning systems.
- **Speculative links.** The broader view of life as stacked timescales and the connections to quantum-like geometry and to consciousness are collected in the Outlook (Section 11). The rest of the programme does not depend on them.
- **The roots, not the whole tree.** The programme aims to explain where the brain's capacities come from, not the full complexity with which a particular brain realizes them (Section 1.4). Success would leave most of neuroscience's hard problems open, while giving them, we hope, a common starting point.
- **Levels and causation.** Whether attractors are causes or descriptions ultimately touches philosophical questions that the programme does not settle (Kim, 1998). We argue only for causal relevance in senses that can be tested: the creation of timescales, the effects of interventions, and closure (Section 5.1; P17).

10. Relation to other approaches

The programme is meant to complement neighbouring approaches rather than compete with them. The table summarizes where each meets it and what the programme adds.

Approach	Central object	Where it meets this programme	What this programme adds
Neuroconnectionism, task-optimized deep networks	Architecture, objective and training data that make networks brain-like	Shared aim: computational models tested against brain and behavioural data	A first-principles route to the learning rule and dynamics; whole-brain generative modelling; continual learning by construction
Biophysical whole-brain models (neural mass, neural field)	Biophysics at one chosen scale	Shared target: large-scale brain dynamics	A scale-agnostic family of rules that complements scale-specific mechanisms; task fidelity, as networks trainable as learners

Approach	Central object	Where it meets this programme	What this programme adds
Data-driven reconstruction of brain dynamics (recurrent networks fitted to recordings)	Generative dynamical models learned from data	Shared goal: generative models of brain dynamics	A model class derived from first principles, with interpretable priors, inputs and learning
Free Energy Principle, active inference	Variational free energy minimization by self-organizing systems	The networks minimize variational free energy at two timescales	Sufficient physical conditions for one minimal system; no Markov blanket required; falsifiable predictions
Bayesian brain, predictive coding	Posterior inference; prediction errors	Prediction errors, posteriors and precisions appear in the construction	A physical origin for prediction errors and precision; the timing of the learning rule
Sampling-based probabilistic computation	Neural variability as samples from a posterior	Fluctuations sample the posterior at two timescales	Variance equals gain; noise as precision; a learned Fisher metric
Hopfield networks, Boltzmann machines, dense associative memories	Energy landscapes; contrastive learning; storage capacity	The networks generalize stochastic Hopfield networks and Boltzmann machines	Local, single-phase learning; asymmetric couplings; self-orthogonalization
Landscape–flux theory of non-equilibrium systems	Decomposition into potential and curl flux	Symmetric and antisymmetric couplings split gradient and solenoidal flow	A learning rule expected to produce this split on-line (self-solenoidization)
Latent-cause models of learning and extinction	Inference over hidden causes of experience	Extinction as the formation of a new cause	An attractor-level mechanism with neural predictions
Physical learning (equilibrium propagation, coupled learning)	Learning rules native to physical networks	Shared aim: learning native to physics	Unsupervised, generative learning derived from first principles; a link to brain data
Self-assembling learning circuits	Learning circuits that assemble themselves from local plasticity and approximate gradient descent	Shared stance: learning should emerge from local rules rather than be designed; a minimal building block, then composition (Liao et al., 2026)	The learning rule itself derived from the physics of the unit; inference with attractor priors rather than supervised gradient descent; recurrent, scale-agnostic networks fitted to brain data
Continual-learning methods in machine learning	Avoiding catastrophic forgetting	Shared benchmarks	Resistance to forgetting as an emergent property, not an add-on
Information-theoretic accounts of emergence (causal emergence; informational and causal closure)	Macroscopic variables that are self-contained, or causally more informative than the micro level	Attractors as candidate closed macro-variables of brain dynamics	A mechanism that generates such levels from within, through learned attractors with their own timescales; an fMRI test (P17)
Connectome-based attractor and energy-landscape models of brain data	Energy landscapes or attractors of Ising- or Hopfield-type models fitted to, or built from, brain connectivity	The same model family and the same data (Watanabe et al., 2013; Ezaki et al., 2017; Gu et al., 2018; Fasoli et al., 2026)	A derived learning rule; attractors read as Bayesian priors; self-orthogonalization as a testable signature (P1, P2)
Evolution and development as learning	Associative learning by networks of interacting components over evolutionary and developmental time	Learning as a property of organized matter, across levels (Watson et al., 2011; Vanchurin et al., 2022; Watson and Levin, 2023; Fields and Levin, 2020, 2022)	A minimal physical construction of the elementary step, with tests in brains and machines; a reading of evolution as refining, rather than inventing, that step

Approach	Central object	Where it meets this programme	What this programme adds
Coarse-graining and renormalization of neural data	How a description changes with resolution	Shared question: what survives coarse-graining (Machta et al., 2013; Meshulam et al., 2019)	A mild, testable condition for parcellated fMRI (homogeneity on the attractors), and a camera model with explicit controls
Stochastic thermodynamics of machine learning	Generative models as non-equilibrium stochastic processes, with free energies, entropy production and fluctuation theorems	Shared tools: free energies, Langevin and score-based sampling (Welling et al., 2026; Goldt and Seifert, 2017; Sohl-Dickstein et al., 2015), the thermodynamic cost of learning, thermodynamic hardware	A local learning rule derived from the same physics; attractor-level predictions for brain data

Eight criteria summarize what the construction is meant to achieve. They are listed, with the sections where each is addressed, in *Desiderata: what the construction should achieve*.

Desiderata: what the construction should achieve

#	Criterion	What it requires	Addressed in
D1	Physically grounded	Inference and learning follow from stated physical assumptions about coarse observables, not from algorithms imposed from outside.	Section 3
D2	Scale-agnostic	Rules of the same family apply, approximately, at any level of description (unit, region, network), so that the model class transfers across scales and coarse measurements such as fMRI are legitimate targets. Ideally, the model also explains how new levels of description arise from within the system.	Sections 4.2, 5
D3	Local	Learning uses only quantities available at the unit: no global loss and no error transport.	Sections 3.3, 4.1
D4	Constructive and generative	The model can be built from its stated rules, as a simulation or in hardware, ideally running natively on physical substrates with noise as a resource. As a brain model, it is not only fitted to data but can also generate data that were not measured, including the effects of counterfactual interventions.	Sections 5.2, 8
D5	Trainable	The same model class learns tasks: sequences, continual learning, inference under uncertainty.	Sections 2, 4.3, 8.3
D6	Probabilistic	Uncertainty is represented natively, at both fast (states) and slow (structure) timescales.	Sections 3.3, 4.4
D7	Falsifiable	Assumptions are explicit, each can fail, and the model makes quantitative predictions with stated failure criteria.	Sections 3.2, 7, 9
D8	Interpretable	Model components map onto computational roles (priors, likelihoods, prediction errors) and onto measurable quantities.	Sections 3.3, 4, 5.2

Intellectual lineage and acknowledgements

This programme stands on many traditions, and most of its ideas are new combinations of theirs rather than independent innovations. We acknowledge in particular:

- **Statistical physics and information geometry:** maximum entropy (Jaynes, 1957), fluctuation–dissipation (Kubo, 1966), fluctuation theorems and work relations (Jarzynski, 1997;

Crooks, 1999; Seifert, 2012), thermodynamic geometry (Sivak and Crooks, 2012), projection operators and memory kernels (Zwanzig, 1961; Mori, 1965), averaging and multiscale methods (Pavliotis and Stuart, 2008), synergetics and slaving (Haken, 1983), centre manifolds (Carr, 1981), non-parametric information geometry (Pistone and Sempi, 1995) and information geometry more broadly (Amari, 1998, 2016).

- **The free energy principle and active inference:** Friston (2010, 2019), Palacios et al. (2020), Parr et al. (2022), Friston et al. (2023a,b), Ramstead et al. (2023), Sakthivadivel (2022) and Friston et al. (2025); and the critical literature that sharpened our formulation (Andrews, 2021; Biehl et al., 2021; Millidge et al., 2021; Aguilera et al., 2022; Bruineberg et al., 2021).
- **The Bayesian brain and predictive coding:** Rao and Ballard (1999), Knill and Pouget (2004), Friston (2005), Feldman and Friston (2010), Bastos et al. (2012), Whittington and Bogacz (2017), Song et al. (2024) and Senn et al. (2024); and hierarchical Bayesian models of learning under uncertainty (Mathys, 2011).
- **Sampling-based probabilistic computation:** Hoyer and Hyvärinen (2003), Fiser et al. (2010), Berkes et al. (2011), Buesing et al. (2011), Kappel et al. (2015), Aitchison and Lengyel (2016), Orbán et al. (2016), Echeveste et al. (2020) and Hénaff et al. (2020); and probabilistic population codes as an alternative (Ma et al., 2006).
- **Attractor networks and energy-based models:** Hopfield (1982), Amit et al. (1985), Ackley et al. (1985), Smolensky (1986), Personnaz et al. (1986), Sompolinsky and Kanter (1986), Kleinfeld (1986), Kanter and Sompolinsky (1987), Amit (1989), Hinton (2002), Barra et al. (2012), Krotov and Hopfield (2016), Ramsauer et al. (2020), Khona and Fiete (2022), Hoover et al. (2025) and Clark (2026); Hebbian unlearning and dreaming (Hopfield et al., 1983; Fachechi et al., 2019; Serricchio et al., 2025); and perturbation evidence for attractor dynamics in brains (Kim et al., 2017; Inagaki et al., 2019; Finkelstein et al., 2021).
- **Local Hebbian and anti-Hebbian learning:** Oja (1982), Földiák (1990), Bi and Poo (1998), Gerstner et al. (2018), Pehlevan and Chklovskii (2019) and Lillicrap et al. (2020).
- **Non-equilibrium dynamics and metastability:** potential–flux decompositions (Ao, 2004; Kwon et al., 2005; Wang et al., 2008; Friston and Ao, 2012; Yan et al., 2013), irreversible samplers (Ma et al., 2015; Rey-Bellet and Spiliopoulos, 2015), metastability and itinerancy (Kelso, 1995; Tsuda, 2001; Rabinovich et al., 2008; Tognoli and Kelso, 2014; Hancock et al., 2024) and broken detailed balance in the brain (Lynn et al., 2021; Kringelbach et al., 2024) and the non-equilibrium thermodynamics of asymmetric networks (Aguilera et al., 2023).
- **Large-scale brain dynamics:** resting-state networks and their relation to tasks (Raichle et al., 2001; Fox et al., 2005; Smith et al., 2009; Cole et al., 2016; Margulies et al., 2016; Dosenbach et al., 2025); whole-brain models (Deco et al., 2008, 2010; Honey et al., 2009; Sanz Leon et al., 2013; Breakspear, 2017); maximum-entropy and energy-landscape models (Schneidman et al., 2006; Watanabe et al., 2013; Ezaki et al., 2017; Gu et al., 2018); connectome-based attractor models of fMRI (Englert et al., 2026; Fasoli et al., 2026); attractor manifolds and timescale separation (Runfola et al., 2025; Athanasiadis et al., 2025); metastable attractor dynamics and expectation (Mazzucato et al., 2019); parcellation and its evaluation (Gordon et al., 2014; Eickhoff et al., 2018; Urchs et al., 2019) and precision functional mapping (Gordon et al., 2017); brain-state models (Vidaurre et al., 2017); data-driven reconstruction of brain dynamics (Koppe et al., 2019; Singh et al., 2020; Durstewitz et al., 2023); and the debate between bottom-up and top-down neuroscience (Rosas et al., 2025).
- **Continual learning:** McCloskey and Cohen (1989), McClelland et al. (1995), French (1999), Kirkpatrick et al. (2017), Shin et al. (2017), Zeng et al. (2019), Hadsell et al. (2020), van de Ven et al. (2020), Libby and Buschman (2021), Flesch et al. (2022) and Chandra et al. (2025).
- **Learning, extinction and expectation:** Rescorla and Heth (1975), Bouton (2004), Rescorla (2004), Redish et al. (2007), Gershman et al. (2010), Milad and Quirk (2012), Büchel et al. (2014), Wager and Atlas (2015) and Tabor et al. (2017).
- **Physical, neuromorphic and thermodynamic computing:** Strukov et al. (2008), Still et al. (2012), England (2015), Prezioso et al. (2015), Goldt and Seifert (2017), Scellier and Bengio (2017), Camsari et al. (2019), Borders et al. (2019), Conte et al. (2019), Hooker (2021),

Stern et al. (2021), Schuman et al. (2022), Coles et al. (2023), Harabi et al. (2022), Stern and Murugan (2023), Stern et al. (2025), Jelinčič et al. (2025) and Welling et al. (2026).

- **NeuroAI:** Hassabis et al. (2017), Lake et al. (2016), Richards et al. (2019), Ali et al. (2022), Doerig et al. (2023) and Zador et al. (2023).
- **Scale-free and basal cognition:** Lyon (2005), Watson et al. (2011), Levin (2019), Fields and Levin (2022), Vanchurin et al. (2022), Watson and Levin (2023), Friston et al. (2025) and Yampolskaya and Mehta (2026).
- **Emergence, levels and causation:** hierarchy and near-decomposability (Simon, 1962), "more is different" (Anderson, 1972), order parameters and circular causality (Haken, 1983; Kelso, 1995), hierarchies of timescales in the brain (Kiebel et al., 2008; Murray et al., 2014) and in recurrent networks (Yamashita and Tani, 2008), interventionist causation (Woodward, 2004), causal emergence and closure (Hoel et al., 2013; Rosas et al., 2020, 2024); and, for the opposing view, the causal exclusion argument (Kim, 1998).

We are grateful to the many colleagues whose work and conversations shaped these ideas. Any errors in combining them are our own.

11. Outlook: further-reaching possibilities

The rest of the programme does not depend on the ideas in this section SPECULATIVE.

We state them because they follow naturally from the construction, and because stating them openly makes it easier to see where it might lead, or where it might break.

Life as stacked timescales. From the point of view of Section 5, life stacks timescales, from molecules and cells to circuits, behaviour, development and evolution, and it appears to do so in a way that largely respects the boundaries between levels (Section 5.1). Hopfield-type attractor dynamics indeed recur across biology, from cell differentiation and epigenetic memory to molecular self-assembly (Yampolskaya and Mehta, 2026); local adaptation of interactions can produce associative memory at the scale of a whole system (Watson et al., 2011); and evolution and development have been cast as learning across nested levels and timescales (Vanchurin et al., 2022; Watson and Levin, 2023). That living systems largely respect these boundaries is a great gift for scientific understanding, and it is what makes the programme's bet plausible: if the same elementary step recurs at each level, understanding it once gives a handle on many.

Evolution as refining a law rather than inventing rules. If the elementary step is a property of organized matter rather than a biological invention (Sections 1.4 and 3.4), evolution may have worked less like an engineer designing rules and more like a craftsman perfecting a material. It would start from structures that align only weakly with their environment and favour substrates in which the same law operates with more headroom: more reliable separation of timescales, wider ranges of coupling, richer architectures. Because better-aligned systems can in turn build better substrates, the gains could compound, so that alignment improves at an accelerating, perhaps exponential, rate over evolutionary time. This echoes views of evolution and development as one scale-free process acting on materials that are already competent (Fields and Levin, 2020; Levin, 2023), and of evolution as learning (Watson and Levin, 2023; Vanchurin et al., 2022). Whether any of this can be made quantitative is open.

Quantum-like geometry. The rotational, solenoidal part of the dynamics (Section 4.5) leads into the territory of tangent and cotangent bundles. On a dually flat statistical manifold, Dombrowski's construction gives the tangent bundle a Kähler structure (Dombrowski, 1962; Molitor, 2012, 2013), the same kind of geometry that underlies the geometric formulation of quantum mechanics (Kibble, 1979; Ashtekar and Schilling, 1999). Closed solenoidal loops may then accumulate geometric, Berry-like phases (Berry, 1984), which are known to arise in classical stochastic kinetics (Sinitsyn and Nemenman, 2007). Whether this formal kinship can account for quantum-like phenomena in cognition and decision making (Busemeyer and Bruza, 2012; Pothos and Busemeyer, 2013) is open.

Consciousness. Attractors offer a plausible latent dimension for a global workspace (Baars, 1988; Dehaene et al., 1998; Mashour et al., 2020; Whyte and Smith, 2021) or for the "inner screen" of Ramstead et al. (2023). We make no claim here beyond pointing out these links.

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